

24. Peroxyacids	980	2-6. Comparison of literature data for enthalpies of vaporization with estimated [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] at 298.15 K	1156
25. Carbonates	982	3-1. Group and group codes for aliphatic hydrocarbons and aliphatic oxygen compounds (86PED/NAY)	1158
26. Linear, branched, cyclic, aromatic amines	982	3-2. Group comparisons for aliphatic hydrocarbons and aliphatic oxygen compounds (86PED/NAY)	1158
27. Imines	992	3-3. Group specificity and values for bonding of $-\text{CH}_2-$ to two carbon atoms in aliphatic hydrocarbons (86PED/NAY)	1158
28. Linear, branched, cyclic, aromatic nitriles	992	3-4. Comparison of enthalpies of formation in the gas phase at 298.15 K (kJ/mol) (alkanes, alkenes, alkynes)	1159
29. Hydrazines	997	3-5. Comparison of enthalpies of formation in the gas phase at 298.15 K (kJ/mol) (alcohols, ethers, ketones, acids)	1159
30. Diazenes	998		
31. Azides	1000		
32. Heterocyclic nitrogen compounds CHN	1001		
33. Linear, branched, cyclic, aromatic amides	1006		
34. Ureas	1011		
35. Amino acids and peptides	1014		
36. Nitroso and cyanato compounds	1021		
37. Linear, branched, cyclic, and aromatic nitro compounds	1022		
38. Nitrites	1031		
39. Nitrates	1032		
40. Nitramines	1033		
41. Cyclic CHNO	1035		
42. Linear, branched, cyclic, and aromatic thiols .	1035		
43. Linear, branched, and aromatic sulfides	1041		
44. Disulfides	1048		
45. Sulfoxides	1049		
46. Linear, branched, and aromatic sulfones	1050		
47. Sulfites	1055		
48. Sulfates	1055		
49. Heterocyclic sulfur CHS compounds	1056		
50. Fluorides, CHF and CHFO compounds	1058		
51. Chlorides, CHCl and CHClO compounds	1066		
52. Bromides, CHBr and CHBrO compounds	1086		
53. Iodides, CHI and CHIO compounds	1092		
54. CHClF , CHClBr , CHBrF , CHFI compounds .	1099		
55. Summary of Residuals for C-H-N-O-S-Halogen Families	1102		
56. Name-Formula-CASRN-Family-Page Index ..	1106		

List of Tables In the Appendices

1-1. Groups derived from data on a single compound	1152
2-1. Comparison of literature data for [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] at 298.15 K and enthalpies of vaporization corrected to 298.15 K with estimated [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] for <i>n</i> -alkanes	1154
2-2. Comparison of literature data for [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] at 298.15 K and enthalpies of vaporization corrected to 298.15 K with estimated [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] for <i>n</i> -alkane thiols ...	1154
2-3. Comparison of literature data for [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] at 298.15 K and enthalpies of vaporization corrected to 298.15 K with estimated [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$] for alkyl sulfides	1154
2-4. Comparison of literature data for enthalpies of fusion with estimated [$\Delta_f H^\circ(\text{solid}) - \Delta_f H^\circ(\text{liq})$] at 298.15 K	1155
2-5. Comparison of literature data for entropies of fusion with estimated [$S^\circ(\text{solid}) - S^\circ(\text{liq})$] at 298.15 K	1155

1. Introduction

The purpose of this paper is to demonstrate that the estimation of thermodynamic properties of organic compounds in the condensed phase at 298.15 K and 101,325 Pa can be carried out in a satisfactory manner using established second-order group-additivity methods. The second-order group-additivity method, originally introduced by S.W. Benson and coworkers (58BEN/BUS, 68BEN, 69BEN/CRU, 69SHA, 71SHA, 73EIG/GOL, 76BEN, 77LUR/BEN, 77SHA/GOL, 77STE/GOL) for estimating the thermodynamic properties of organic compounds, was developed and used primarily for the gas phase. This work includes a re-examination of the capabilities for estimation of the thermodynamic properties of the gas phase to maintain continuity with the condensed phase and also to introduce changes in group values necessitated by more recent thermodynamic data available in the literature.

The coverage of organic compounds includes those substances containing the elements: carbon, C; hydrogen, H; oxygen, O; nitrogen, N; sulfur, S; fluorine, F; chlorine, Cl; bromine, Br; and iodine, I. The particular thermodynamic properties for which groups and group values have been determined are: enthalpy of formation, $\Delta_f H^\circ$; heat capacity, C_p° ; and entropy, S° . The entropy of formation ($\Delta_f S^\circ$), Gibbs energy of formation ($\Delta_f G^\circ$), and the natural logarithm of the equilibrium constant ($\ln K_f$) for the formation reaction, are calculated as auxiliary properties.

The second-order group-additivity approach has been generally accepted by physical chemists and chemical engineers because of the simple basis of additivity, clarity of notation, second-order character, i.e., inclusion of nearest-neighbor interactions, ease of application, and satisfactory agreement between the thermodynamic value reported in the literature and its estimated value. The ASTM Chemical Thermodynamic and Energy Release Program, CHETAH, (74SEA/FRE)) uses these methods for the estimation of the thermodynamic properties of organic compounds in the gas phase and for the classifi-

cation of chemicals or compositions depending upon whether they are likely to be impact sensitive. The AIChE Design Institute for Physical Property Data (DIPPR) Manual for Predicting Chemical Process Design Data (83DAN/DAU) recommends the second-order group-additivity method (76BEN) for the estimation of enthalpy of formation, heat capacity, and entropy of organic compounds at 298.15 K in the gas phase. DIPPR Project 871 is an experimental project begun to determine accurate enthalpies of formation and vaporization of key organic compounds so that reliable gas phase enthalpies of formation can be calculated at 298.15 K. The latter data would then permit either the calculation of new second-order group-additivity values or the replacement of group values which are based on poor quality data. The improved or new group values would be used to upgrade both the DIPPR Data Prediction Manual and the ASTM CHETAH Program.

With such broad needs for thermodynamic property estimation in ASTM and AIChE, we felt that the successful application of this approach for the gas phase merited an intensive examination of its application into the condensed phase. The approach taken for the evaluation of thermodynamic data and the path of development of groups and group values for hydrocarbon compounds came from our previous paper (88DOM/HEA) in which a systematic procedure was followed for the selection of group values which gave minimum residuals between the literature and calculated values. The *n*-alkanes from C₂ to C₁₈ make up one of the most studied families of compounds and have some of the most pristine values for $\Delta_f H^\circ$, C_p° , and S° of all the families of organic compounds. They form the basis for the development of the C-(H)₃(C) and C-(H)₂(C)₂ group values. From this point, the group and group value development proceeds to branched alkanes, alkenes, alkynes, aromatic and alicyclic compounds, CHO compounds (alcohols, ethers, etc.), CHN compounds (amines, nitriles, etc.), and on to CHNO, CHS, CHSO, and organic halogen compounds. Care is needed in the development of group values because experimental or recommended data have different degrees of quality and are not homogeneous. Reduction of data to as common a basis as possible is required. For combustion data, some previously established guidelines were used (71DOM). A global least squares, least sums, or regression-type fit of all the group values was not performed because of the differences in the quality of the data, and because of the limited amount of data available for the generation of certain groups and group values. The generation of groups and the calculation of group values was in part manual and in part computer-assisted. Some computations for average values, average deviations, or standard deviation were performed using a desk-top calculator. Others were made using computer spread-sheet analysis.

The group values generated for the hydrocarbons were held fixed for the generation of non-hydrocarbon values. Most of the group values for non-hydrocarbon compounds were generated using the THERM/EST Program

(NIST Standard Reference Database 18) (90DOM/HEA2) by having the group value being sought initially become part of the residual value. In this procedure, an unknown group value is calculated, then a value is assigned to the unknown group, and a final or true residual value is calculated which excludes the new group value, but minimizes the final residual value. When a large number of experimental values were available to calculate a group value, as with *n*-alkanes, *n*-alkanols or *n*-alkanethiols, final adjustments were examined with computer spread-sheet software. Care was taken to accommodate the adjusted group values when the same groups were needed for different families of compounds. For example, the C-(H)₂(C)(O) group and its group values are required for alcohols, ethers, esters, and peroxides. Group values for some families or compounds were generated simply by calculating simple arithmetic averages because the experimental data were limited to 2 or 3 values. The group C-(H)₃(C) is used freely in the molecular structure representation of compounds because the value of the methyl group does not change except for the physical phase no matter to what it is attached. Hence, C-(H)₃(C) = C-(H)₃(O) = C-(H)₃(N) = C-(H)₃(S), etc., and consequently, methanol can be represented by: C-(H)₃(C) + O-(H)(C) rather than: C-(H)₃(O) + O-(H)(C). Appendix 1 lists unique groups derived from data on individual compounds as their source.

The compounds for which estimated properties have been calculated are divided into various organic families as shown in Table 1. The number of compounds within each family is indicated. A total of 1512 compounds have had thermodynamic properties estimated and compared with a literature value. This total is not exhaustive, but does represent a sufficiently broad array of organic structures to demonstrate the applicability of the group additivity method to the condensed phase. Compounds are listed according to the increasing number of carbon atoms within each family, but sometimes the carbon number reverts back to lower values because of the inclusion of certain compounds with secondary or tertiary substitution, unsaturation, multiple functional groups, aromatic substitution, or cyclic structures. Table 2 provides a listing of over 600 groups and energy corrections, and their corresponding values for $\Delta_f H^\circ$, C_p° , and S° for the gas, liquid, and solid phases. Table 3 offers some guidance to the definition and interpretation of organic groups for persons not accustomed to the notation in Table 2.

Thermodynamic properties for 1512 compounds are shown in Tables 4 through 54 and contain the following information: the title of the organic family, the number of compounds included in that family, individual compound name(s), formula of the compound, the organic groups which comprise the structure of the compound, symmetry number, σ , and optical isomers, η . Thereafter, each phase is treated separately, gas, liquid, and solid. Property symbols $\Delta_f H^\circ$ (in kJ/mol), C_p° (in J/mol·K), and S° (in J/mol·K) appear in the extreme left column. Next, the ex-

perimental or recommended values are also given, when available, and entered under the column, "Literature". The property value derived from summing the group values is given under the column, "Calculated", so that a difference or residual value may be calculated and shown under the column, "Residual". The residual offers an indication of how close the estimated value comes to the one determined experimentally or to the one derived from an evaluation of a collection of literature data. If the appropriate group values are available so that the enthalpy of formation and entropy can be estimated, we also provide the entropy of formation, $\Delta_f S^\circ$ (in J/mol·K), Gibbs energy of formation, $\Delta_f G^\circ$ (in kJ/mol⁻¹), and the logarithm of the equilibrium constant for the formation process, $\ln K_f$. In order to calculate the entropies of formation at 298.15 K and 101,325 Pa, the entropies of the elements in their standard states are needed as well as the entropies of the respective compounds. The entropies of the elements at 298.15 K and at 101,325 Pa have been obtained from (89COX/WAG) and are as follows in J/mol·K: carbon (cr, graphite), 5.740; H₂ (gas), 130.571; O₂ (gas), 205.043; N₂ (gas), 191.500; S (cr, rhombic), 32.054; F₂ (gas), 202.682; Cl₂ (gas), 222.972; Br₂ (liquid), 152.21; I₂ (cr), 116.14. Estimation of the entropy in the gas phase requires a $-R \ln \sigma$ term where σ is the total symmetry number of the molecule and R is the universal gas constant ($R = 8.31451$ J/mol·K). The total symmetry number of the molecule is divided into two parts: the internal symmetry number and the external symmetry number. The definitions of the latter symmetry numbers and several example calculations of symmetry numbers can be found in (88DOM/HEA). Estimation of the gas phase entropy also requires accounting for optical isomers as the molecular structure dictates with a $R \ln \eta$ term, where η indicates the number of such isomers. Since this work deals with data at only one temperature, 298.15 K, no distinction is made between the heat capacity at constant pressure, C_p° , and the saturation heat capacity, C_{sat} .

Reference squibs appear in the extreme right column of the tables under the heading "References"; The squibs are of the type XXAAA/BBBN for each property in each of the phases and correspond to entries under the column "Literature". In this squib notation, XX denotes the last two digits of the year of publication of the paper, AAA, the first three letters of the last name of the first author, and BBB, those of the last name of the second author (if present). Authors after the first two are not noted. The numeral, N, at the end of the squib is present only when the authors have more than one paper published in a given year. Table 55 provides a summary of the residuals for each family which offers some global insight into the agreement realized between literature and calculated values. The residuals have been divided into those for $\Delta_f H^\circ$ which were $< \pm 4$, $> \pm 4$ but $< \pm 8$, and $> \pm 8$ kJ/mol⁻¹. Similarly, for C_p° and S° , the residuals have been divided into those which were $< \pm 4$, $> \pm 4$ but $< \pm 8$, and $> \pm 8$ J/mol·K. Table 56 shows an alphabetical compound name-formula index which provides the CAS registry number, family in which the compounds may be found, its

listing or rank within the given family, and the page on which data for the compound appears. A bibliography given in Table 57 with about 1000 references links reference squibs in Tables 4 through 54 to literature citations.

We have examined the original reference sources for data on the enthalpies of formation, heat capacities, and entropies for almost all of the compounds. The thermodynamic tables compiled by Stull, Westrum, and Sinke (69STU/WES, 69STU/WES2) have been used for many of the literature gas phase heat capacities and entropies. We have also used some general thermodynamic reference sources to find original sources in certain cases (70COX/PIL, 71ZWO/WIL, 72DOM, 77PED/RYL, 84DOM/EVA, 85MAJ/SVO, 86TRC, 86TRC2, 86PED/NAY, 90DOM/HEA). Over 3700 comparisons between literature and calculated values are shown for $\Delta_f H^\circ$, C_p° , and S° in the gas, liquid, and solid phases. Approximately half of the comparisons are for the condensed phase.

Interpretation is occasionally required when a reference squib is designated for a specific property. For example, a reference squib denoted for $\Delta_f H^\circ$ in the gas phase may not actually provide that specific property, but will report an enthalpy of vaporization at 298.15 K which when added to $\Delta_f H^\circ$ in the liquid phase, will then be equal to the designated experimental or recommended $\Delta_f H^\circ(\text{gas})$ value. Similarly, a reference squib denoted for $\Delta_f H^\circ(\text{solid})$ may not contain the actual $\Delta_f H^\circ$ property for the solid phase, but does report the ΔH° for the fusion or melting of the compound. When the latter is corrected from the melting temperature to 298.15 K and combined with $\Delta_f H^\circ(\text{liq})$ at 298.15 K, one obtains the $\Delta_f H^\circ(\text{solid})$ value. Despite concerns related to the estimation of thermodynamic properties for solid substances, we typically find good agreement between literature and calculated values. Common doubts about the ability to develop a predictive scheme for solid substances arise because some organic compounds have many crystalline forms in the proximity of the melting point. The second-order group-additivity approach does have its limits. We expect that the predicted solid phase at 298.15 K is the same as the stable form encountered experimentally at 298.15 K. For organic compounds with multiple crystalline phases and solid-solid transitions, as found with the C₁₃, C₁₄, and C₁₅ 1-alkanols, the group additivity approach provides only a limited value for the thermodynamic property. We do not feel that this limitation diminishes the overall usefulness of prediction of this method for solid organic substances. Appendix 2 shows that internal consistency does exist when comparisons are made between literature values for enthalpies and entropies of fusion and enthalpies of vaporization and the estimated differences for [$\Delta_f H^\circ(\text{solid}) - \Delta_f H^\circ(\text{liq})$], [$S^\circ(\text{solid}) - S^\circ(\text{liq})$], and [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$], at 298.15 K. Differences between literature values for enthalpies and entropies of fusion corrected from the melting temperature to 298.15 K and [$\Delta_f H^\circ(\text{solid}) - \Delta_f H^\circ(\text{liq})$] and [$S^\circ(\text{solid}) - S^\circ(\text{liq})$] yield average deviations of ± 2.7 kJ/mol⁻¹ and ± 4.7 J/mol·K, respectively. A similar comparison between literature values for the enthalpy of vaporization corrected to 298.15 K

and $[\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})]$ gives an average deviation of $\pm 1.6 \text{ kJ}\cdot\text{mol}^{-1}$.

The quality of the groups and group values can be evaluated by examining the magnitude of the difference (or residual) which is observed between the literature and calculated values for a specific property in a given physical phase. For $\Delta_f H^\circ$, differences within $\pm 4 \text{ kJ/mol}$ constitutes very good or satisfactory agreement, those which are between ± 4 and $\pm 8 \text{ kJ/mol}$ are at the limits of acceptability, and differences which are greater than $\pm 8 \text{ kJ/mol}$ are symptomatic of a problem. The occurrence of differences larger than $\pm 8 \text{ kJ/mol}$ are usually due to poor quality literature data or to a neglected molecular interaction, both of which can lead to the incorrect assignment for a group value. A similar situation applies to heat capacity and entropy differences. When differences within $\pm 4 \text{ J/mol}\cdot\text{K}$ occur, the agreement is considered very good, when they are between ± 4 and $\pm 8 \text{ J/mol}\cdot\text{K}$, the agreement is acceptable, and when the differences are greater than $\pm 8 \text{ J/mol}\cdot\text{K}$, they reflect a problem, which similarly can be related to poor quality data or to a neglected molecular interaction, and can lead to the selection of an incorrect group value.

Certain molecules such as methane, methanal, acetonitrile, nitromethane, and the methyl halides are precluded from the rules of group additivity because they are structurally comprised of only one group and, hence, their group value is equivalent to the corresponding property value in each of the phases. We have included a number of such substances at the beginning of some of the organic families for comparison purposes. When needed for various calculations, the 1989 table of atomic weights was used (91DEL/HEU).

2. Discussion of Results

2.1. Hydrocarbon Compounds

The hydrocarbon compounds and thermodynamic properties appearing in 88DOM/HEA are also presented here as well as the calculation of the entropy of formation, Gibbs energy of formation, and equilibrium constant for the formation reaction. A total of 48 hydrocarbon compounds has been added which have created more groups and group values. A small number of groups and group values appearing in 88DOM/HEA have also been modified. Hydrocarbons comprise the most studied single family of organic compounds from a thermodynamic standpoint and form the foundation for the development of groups and group values not only within hydrocarbons themselves but also for non-hydrocarbon compounds. Thermodynamic property comparisons between hydrocarbons and non-hydrocarbons permit one to test whether additivity is being preserved, whether molecular forces are interacting, or whether the literature values may be suspect.

The hydrocarbon compounds examined have been divided into eleven families: *n*-alkanes, *t*-alkanes, *q*-alkanes, *n*-alkenes, *s*-alkenes, alkynes, aromatic

CH-01, aromatic CH-02, cyclic CH-01, cyclic CH-02, and cyclic CH-03. These families contain thermodynamic property estimates for a total of 427 hydrocarbon compounds and are found in Tables 4 through 14. An examination of the 532 comparisons between literature and calculated values for $\Delta_f H^\circ$ shows that 70 percent of the residuals are $< \pm 4 \text{ kJ/mol}$, 17 percent are $> \pm 4$ but $< \pm 8 \text{ kJ/mol}$, and 13 percent are $> \pm 8 \text{ kJ/mol}$. For C_p° with 361 comparisons, we find 85 percent of the residuals $< \pm 4 \text{ J/mol}\cdot\text{K}$, 8 percent $> \pm 4$ but $< \pm 8 \text{ J/mol}\cdot\text{K}$, and 7 percent $> \pm 8 \text{ J/mol}\cdot\text{K}$. Similarly, for S° with 338 comparisons, we find 76 percent of the residuals $< \pm 4 \text{ J/mol}\cdot\text{K}$, 16 percent $> \pm 4$ but $< \pm 8 \text{ J/mol}\cdot\text{K}$, and 8 percent $> \pm 8 \text{ J/mol}\cdot\text{K}$.

A novel approach for dealing with branched alkanes has been described in (88DOM/HEA) for tertiary and quaternary carbon atoms in hydrocarbon compounds. It corrects for the repulsive interactions of hydrogen atoms on methyl groups attached to tertiary or quaternary carbon atoms and improves the agreement between literature and estimated values. The corrections for methyl repulsion in branched hydrocarbons have been developed only for $\Delta_f H^\circ$ at 298.15 K. It accommodates the observation that as branching increases for an isomeric hydrocarbon, the $\Delta_f H^\circ$ value becomes more negative (e.g., $\Delta_f H^\circ$'s for isomeric pentanes). A summary of this approach can be found in (88DOM/HEA).

Except for *n*-hexacosane, residuals calculated from literature and calculated values for C_p° and S° for *n*-alkanes with carbon atoms C_{20} and higher are large, but do not come from recent calorimetric investigations. We suggest that some re-determination and confirmation is needed for the C_p° and S° values for these hydrocarbon compounds.

2.2. Organic Oxygen Compounds

After hydrocarbon compounds, organic oxygen compounds are the next most abundant category of organic substances for which thermodynamic data are available. The CHO compounds have been divided into 11 families: alcohols, ethers, aldehydes, ketones, acids, anhydrides, esters, peroxides, hydroperoxides, peroxyacids, and carbonates, and are found in Tables 15 through 25. These tables contain thermodynamic property estimates for 381 CHO compounds. An examination of 570 comparisons of literature and calculated values for $\Delta_f H^\circ$ shows that 62 percent have residuals which are $< \pm 4 \text{ kJ/mol}$, 18 percent are $> \pm 4$ but $< \pm 8 \text{ kJ/mol}$, and 20 percent are $> \pm 8 \text{ kJ/mol}$. Residuals for C_p° show that 72 percent are $< \pm 4 \text{ J/mol}\cdot\text{K}$, 15 percent are between $> \pm 4$ and $< \pm 8 \text{ J/mol}\cdot\text{K}$, and 13 percent are $> \pm 8 \text{ J/mol}\cdot\text{K}$. For S° , 72 percent of the residuals are $< \pm 4 \text{ J/mol}\cdot\text{K}$, 16 percent are $> \pm 4$ but $< \pm 8 \text{ J/mol}\cdot\text{K}$, 12 percent are $> \pm 8 \text{ J/mol}\cdot\text{K}$.

Comparison of literature and calculated values for $\Delta_f H^\circ$, C_p° , and S° shows that for primary alcohols the agreement is reasonably good. However, initial agreement between literature and estimated values for secondary and tertiary alcohols was not as good as with

primary alcohols. We found that significantly smaller residuals resulted for secondary and tertiary alcohols if a methyl repulsion correction was applied for tertiary or quaternary carbon atom attached to an oxygen atom.

Agreement between literature and calculated values for secondary aliphatic alcohols, diols, triols, and tetrols for $\Delta_f H^\circ$, C_p° , and S° are somewhat inconsistent. Large residuals occasionally appear but do not seem to show a consistent trend. The residuals for some phenolic compounds can be improved with the application of an *ortho* correction for OH–OH interactions. Some of the literature data are not recent and may be suspect.

We found that better agreement between literature and calculated values was obtained if separate C–(H)(C)2(O) and C–(C)3(O) groups were assigned to alcohols and peroxides, and another for ethers and esters, rather than having global groups for all of the organic oxygen families. Hence, this separation is indicated in the list of groups and group values in Table 2 and under each compound in Tables 4 through 54 in the structural group notation.

Literature and calculated values for $\Delta_f H^\circ$, C_p° , and S° for ethers and ketones show generally good agreement. This is possible because of a significant quantity of good quality data in the gas and condensed phases.

For aldehydes, agreement between literature and calculated values for $\Delta_f H^\circ$, C_p° , and S° in the gas phase are generally satisfactory. Although satisfactory agreement is found for $\Delta_f H^\circ(\text{liq})$, agreement for C_p° and S° in the liquid phase is poor. The C_p° and S° data of 56PAR/KEN at 298.15 K for butanal and heptanal in the liquid phase reflect the expected linear relationship when *n*-alkanals increase by a CH₂ group. The recent data reported for ethanal by 88LEB/VAS, propanal by 77KOR/VAS, butanal by 89VAS/LEB, hexanal by 91VAS/BYK, and heptanal by 83DYA, 84VAS/PET indicate that the relationship for 1-alkanals in the liquid phase is neither linear nor smooth. These authors describe anomalies in the liquid phase which they have found to be due to association in 1-alkanals through hydrogen bonds. The group values we have chosen are based on linearity, hence, significant deviation are reflected in the large residuals which occur.

We have found some large differences between the literature and calculated values for $\Delta_f H^\circ$ and C_p° for dibasic acids in the gas and solid phases. It is not clear whether these residuals are due to hydrogen bonding in dibasic acids, odd-even carbon atom effects, the need for a group correction factor, poor experimental data, or most of the above. A significant amount of these data were reported in the 1920's. It may be possible that the odd-even relationship which is observed for the melting temperatures of dicarboxylic acids is similarly reflected in their thermodynamic properties. A large fraction of the residuals in $\Delta_f H^\circ$'s for the dibasic acids are $> \pm 8$ kJ/mol.

The experimental $\Delta_f H^\circ$ values for 1-naphthoic and 2-naphthoic acid differ from each other by 9.4 kJ/mol in the gas phase and 12.4 kJ/mol in the solid phase. Examination of the structures of these acids by 74COL/ROU indicates that 2-naphthoic acid is planar, but 1-naphthoic

acid is twisted 11° out of the naphthalene plane due to overcrowding, hence, these structural differences account for the observed energy differences.

Corrections have been developed for *ortho* and *meta* interactions between two or more carboxylic acid groups in aromatic acids. Similar corrections were developed for interactions between methoxy and carboxylic acid groups. In some instances, a clear interaction correction was not developed because the nature of the interaction between adjacent or near-adjacent groups could not be interpreted clearly, and/or the quality of the experimental data did not allow an interpretation. In these cases, we applied the *ortho* and *meta* corrections developed for hydrocarbon compounds.

Difficulty in resolving the agreement between literature and calculated $\Delta_f H^\circ$ values for benzoic anhydride with aliphatic anhydrides led to the development of separate groups for O–(CO)2 and corresponding attachments to aliphatic and aromatic substituents. They are listed in Table 2 as: O–(CO)2, aliphatic and O–(CO)2, aromatic.

Thermodynamic property data on peroxides, hydroperoxides, and peroxyacids are limited to enthalpies of formation. From time to time, agreement between $\Delta_f H^\circ$ experimental and calculated values is poor. This situation is due to a lack of high quality data on these substances and is understandable because of their explosive and thermally sensitive character. The preparation of sufficient amounts of high purity samples of thermally sensitive substances places a very high demand upon any research effort.

The unusually large difference between the literature and calculated $\Delta_f H^\circ$ for diacetyl peroxide (DAP) in the liquid phase (see Table 21, 38.66 kJ/mol) may be due to the instability of the compound. Because of its instability, bomb calorimetric experiments on diacetyl peroxide were performed on a toluene solution (37.53 wt% DAP, 62.47wt% toluene; 57JAF/PRO). Bomb calorimetric experiments were made at only one concentration of DAP in toluene, hence, dilution errors or analytical errors cannot be easily detected. Examination of the experimental and calculated values for dipropionyl and dibutyl peroxides shows their residuals to be satisfactory.

The differences found between the literature and estimated values for peroxy acids are large. The groups developed for the family appears to be the best that can be assembled. If some re-determinations of the thermodynamic properties for peroxy acids can be made, smaller residuals may result.

2.3. Organic Nitrogen Compounds

Literature and estimated thermodynamic properties on organic nitrogen compounds have been divided in seven CHN families and nine CHNO families. The families which comprise the CHN compounds are: amine, imines, nitriles, hydrazines, diazenes, azides, and cyclic CHN compounds and are found in Tables 26 through 32. A total of 137 CHN compounds are shown. Agreement between experimental and calculated values shows 84

percent of the residuals for $\Delta_f H^\circ$ to be $< \pm 4$ kJ/mol, 10 percent of the residuals for $\Delta_f H^\circ$ to be $< \pm 4$ kJ/mol, 10 percent are $> \pm 4$ but $< \pm 8$ kJ/mol, and 6 percent are $> \pm 8$ kJ/mol. For C_p° , 85 percent of the residuals are $< \pm 4$ J/mol·K, 7 percent are $> \pm 4$ but $< \pm 8$ J/mol·K, and 8 percent are $> \pm 8$ J/mol·K. With respect to S° , 77 percent of the residuals are $< \pm 4$ J/mol·K, 15 percent are $> \pm 4$ and $< \pm 8$ J/mol·K, and 4 percent are $> \pm 8$ J/mol·K.

The families which comprise the CHNO compounds are: amides, ureas, amino acids, nitroso, nitro, nitrites, nitrates, nitramines, and cyclic CHNO compounds. A total of 171 CHNO compounds are shown in Tables 33 through 41. Comparison of literature and calculated values show that for $\Delta_f H^\circ$ residuals, 68 percent are $< \pm 4$ kJ/mol, 11 percent are $> \pm 4$ but $< \pm 8$ kJ/mol, and 21 percent are $> \pm 8$ kJ/mol. For C_p° , 80 percent of the residuals are $< \pm 4$ J/mol·K, 6 percent are $< \pm 4$ but $> \pm 8$ J/mol·K, and 14 percent are $> \pm 8$ J/mol·K. For S° , 69 percent are $< \pm 4$ J/mol·K, 10 percent are $< \pm 4$ but $> \pm 8$ J/mol·K, and 21 percent are $> \pm 8$ J/mol·K.

From an initial examination of the differences between the literature and calculated values, the CHN family appears to be amenable to prediction. We have applied the $-\text{CH}_3$ quaternary correction for alkane branching to nitrogen atoms in tertiary amines, *N,N*-dimethylsubstituted amides, and *N,N*-dimethylsubstituted ureas because better agreement resulted between experimental and estimated values. A corresponding application of the $-\text{CH}_3$ tertiary correction to nitrogen atoms in secondary amines, *N*-methylsubstituted amides, or *N*-methyl substituted ureas was not used because it did not lead to significantly smaller differences between experimental and estimated values.

Comparison of the experimental $\Delta_f H^\circ$ for the solid phase of acetamide with the estimated value shows a difference of -10.41 kJ/mol. This difference is larger than one would like. However, anomalous behavior has been reported for crystalline acetamide due to its tendency to supercool (86EMO/NAU). Acetamide forms an unstable solid phase along with a stable form. The stable and unstable forms have melting temperatures of 353.5 K and 342.15 K, and enthalpies of melting of 15.6 and 12.5–12.9 kJ/mol, respectively.

The estimation of the thermodynamic properties of amino acids and peptides in the solid phase is a particularly challenging task. Some amino acids have been the subject of a significant amount of calorimetric study; glycine and hippuric acid are examples. Other amino acids as well as peptides have received only limited calorimetric attention. There is a mixture of high quality, medium quality, and limited quality data on these compounds. Other challenges included accounting for the dipolar nature of amino acids, and identifying differences when possible between (DL) racemic and optically active (R or D, and S or L) isomers. Because the enthalpy of combustion and formation of glycine has been so frequently determined, we have used the experimental values for this amino acid and the corresponding data on

glycylglycine as the basis for deriving the $\text{C}-(\text{H})_2(\text{CO})(\text{N})$ group and group value and the energy correction for the dipolar nature or zwitterion character of aliphatic amino acids. Since the zwitterion nature of amino acids and peptides is a unique property and not prominent in the other organic nitrogen compounds treated in this paper, a separate identification and energy correction was warranted. The establishment of the zwitterion energy correction also allows the $\text{C}-(\text{H})_2(\text{CO})(\text{N})$ group to have property values not seriously divergent from those groups such as $\text{C}-(\text{H})_2(\text{C})_2$, $\text{C}-(\text{H})_2(\text{C})(\text{CO})$, and $\text{C}-(\text{H})_2(\text{C})(\text{N})$. The zwitterion energy correction for solid aliphatic amino acids and peptides for $\Delta_f H^\circ$, C_p° , and S° are -55.10 kJ/mol, -44.50 J/mol·K, and -13.40 J/mol·K, respectively. Using similar reasoning, a zwitterion energy correction was developed for amino acids and peptides containing an aromatic ring, but required differentiation between situations in which a $-\text{CH}_2-$ group breaks the conjugative nature of the aromatic ring from its linkage to the α -carbon of an amino acid or peptide. For these cases, the zwitterion energy is designated as "aromatic I" and was derived from $\Delta_f H^\circ(\text{solid})$ data for phenylalanine and phenylalanine peptides; the zwitterion energy correction (for aromatic I) for $\Delta_f H^\circ$, C_p° , and S° are -32.00 kJ/mol, -20.50 J/mol·K, and -13.00 J/mol·K, respectively. For situations in which the aromatic ring of an amino acid or peptide is bonded directly to amino groups or to carboxylic acid groups where the influence to ring conjugation should be stronger than it is for phenylalanine derivatives, a second aromatic zwitterion energy correction was derived to accommodate the estimation of the aminobenzoic acids, hippuric acid, and hippurylglycine, and designated as "aromatic II"; the zwitterion energy correction (for aromatic II) for $\Delta_f H^\circ$, C_p° , S° are -11.00 kJ/mol, 5.00 J/mol·K, and -9.00 J/mol·K, respectively. These group values are also found in Table 2. Agreement between literature and estimated values is variable. Future reconciliation of the large residuals may result from more precise calorimetric determinations of certain amino acids and peptides as well as some re-adjustment of group values. Better agreement was obtained between literature and estimated values when $\text{N}-(\text{H})_2(\text{CO})$ and $\text{N}-(\text{H})(\text{C})(\text{CO})$ groups were developed separately for amides and ureas in comparison to amino acids and peptides. At this time, it is not clear whether the better agreement is a function of differences in the molecular structure of these organic families or whether more accurate experimental data will offer new changes to their estimation.

The residual value for $\Delta_f H^\circ(\text{solid})$ for nitrosobenzene is large, -85.65 kJ/mol. The group, $\text{CB}-(\text{NO})$, and corresponding $\Delta_f H^\circ$ group values, were derived from experimental data on 4-nitroso-1-naphthol. The $\Delta_f H^\circ$ experimental data for 4-nitroso-1-naphthol (68HAM/FAG) are more reliable than those for nitrosobenzene (30DRU/FLA). The large residual for nitrosobenzene is probably due to either sample purity or to difficulties with experimental bomb calorimetric procedures, or both.

As the development of the second-order group additivity approach to the estimation of thermodynamic properties (57BEN/BUS, 69BEN/CRU) began, the enthalpy of formation of the benzene (C_6H_6) molecule was divided by six to derive the group value for $\Delta_f H^\circ C_B-(H)(C_B)_2$ group in the gas phase. In this division, the resonance or conjugation energy of benzene ($\sim 150 \text{ kJ}\cdot\text{mol}^{-1}$) had also undergone a corresponding division. We have attempted to extend this concept to pyridine and have introduced the $N_1-(C_B)$ group which not only includes the energy content for a property, but also the corresponding component of the conjugation energy which resides in the pyridine molecule. Reasonable success was achieved for pyridine and substituted pyridines in Table 32. The further extension of this concept to five-membered ring systems becomes a more difficult task when they possess an intrinsic and sizeable conjugation energy, but in addition, contain a significant amount of strain energy. This situation is true for five-membered systems such as furan, pyrrole, and thiophene. We have treated the carbon atoms in these five-membered ring systems as benzene carbon atoms, using the $C_B-(H)(C_B)_2$ group, because their conjugation energies are in the range from 65 to $120 \text{ kJ}\cdot\text{mol}^{-1}$. As a result of this treatment, the $\Delta_f H^\circ$ ring strain corrections (rsc) for furan, pyrrole, and thiophene appear as negative values. In contrast, the conjugation energy in 1,3-cyclopentadiene is small when compared with its ring strain energy and its structural description has been assembled using $C_d-(H)(C_d)$ and $C_d-(H)(C)$ groups rather the $C_B-(H)(C_B)_2$ group.

2.4. Organic Sulfur Compounds

The families which comprise the CHS and CHSO compounds are: thiols, sulfides, disulfides, sulfoxides, sulfones, sulfites, sulfates, and cyclic CHS compounds. A total of 138 CHS and CHSO compounds are shown in Tables 42 through 49. Agreement between literature and calculated values shows 80 percent of the residuals for $\Delta_f H^\circ$ to be $< \pm 4 \text{ kJ}\cdot\text{mol}^{-1}$, 14 percent of the residuals to be $> \pm 4$ but $< \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$, and 6 percent to be $> \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$. For C_p° , 92 percent of the residuals are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 7 percent are $> \pm 4$ but $< \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, and 1 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. For S° , 87 percent of the residuals are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 7 percent are $> \pm 4$ but $< \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, and 6 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$.

Excluding hydrocarbon compounds, organic sulfur compounds containing the elements C, H, and S stand out as offering an extremely high quality array of experimental thermodynamic values for $\Delta_f H^\circ$, C_p° , and S° . The establishment of this high quality array of data on CHS compounds is due to the need of the petroleum industry to understand the thermochemistry of organic sulfur compounds because of their presence in petroleum and because of the need to understand their energetics and equilibrium properties in petroleum refining. Much of the effort to establish high quality thermodynamic data for this class of organic compounds resulted from an ex-

perimental effort which took place at the U.S. Bureau of Mines Thermodynamics Laboratory in Bartlesville, OK (now called the National Institute for Petroleum and Energy Research (NIPER)) and at the Thermochemical Laboratory at Lund University, Lund, Sweden. The development of a high precision rotating bomb calorimeter was a key accomplishment which has led to the determination and publication of extremely precise and accurate thermodynamic properties for CHS compounds. A rotating-bomb calorimeter is needed because the final state of sulfur as an aqueous sulfuric acid solution is not homogeneous in its dispersal throughout the interior of the static combustion bomb and is energetically uncertain. An important phase of the research effort focused on the establishment of the enthalpies of formation of aqueous sulfuric acid in various states of dilution. These $\Delta_f H^\circ$'s were then applied toward the identification of the final state of sulfur in the bomb combustion process for organic sulfur compounds. This feature is important because the energetics of the final thermodynamic state of the combustion process must be clearly and precisely defined. Without a knowledge of the final state of sulfur in the form of an aqueous sulfuric acid solution for the combustion reaction, highly precise and accurate data would not be available. For additional information, the reader should examine 56ROS, 62SKI, and 79SUN/MAN.

The research effort in the two laboratories at Bartlesville and Lund was responsible for publication of high quality data available on organic sulfur compounds in the chemical literature and the subsequent good agreement found here between experimental and estimated values as presented in Tables 42 through 49.

Collectively, the residuals shown for thiols and sulfides are very small. Of the organic sulfur families, sulfones appear to be less well-behaved, but agreement here between experimental and estimated values is still reasonably good.

2.5. Organic Halogen Compounds

The families which comprise the CHX and CHXO compounds are: fluorides, chlorides, bromides, iodides, and mixed halogen compounds. A total of 258 halogen compounds are shown in Tables 50 through 54. Agreement between experimental and calculated values shows 54 percent of the residuals for $\Delta_f H^\circ$ to be $< \pm 4 \text{ kJ}\cdot\text{mol}^{-1}$, 17 percent to be $> \pm 4$ and $< \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$, and 29 percent to be $> \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$. For C_p° , 76 percent of the residuals are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 11 percent are $> \pm 4$ but $< \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$, and 13 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. With respect to S° , 70 percent are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 17 percent are $> \pm 4$ but $< \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, and 13 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$.

In contrast to hydrocarbons and organic sulfur compounds, the thermodynamic properties of organic halogen compounds are collectively not known as precisely. In addition, halogen-halogen interactions operate which require interpretation. When these interactions

are overlooked, they tend to make the differences between literature and estimated values larger than they should be.

The use of a rotating bomb calorimeter for the determination of enthalpies of combustion and for the derivation of enthalpies of formation is needed for organic halogen compounds. The enthalpy of formation of the final state of the hydrohalogen acid in aqueous solution must be determined in order to have a defined thermodynamic final state for the combustion process. Additional problems prevail with organic chlorine compounds in that they form about 15–20% elemental gaseous chlorine, Cl_2 , and about 80–85% HCl in aqueous solution during the bomb combustion. A reducing agent such as a solution of arsenious oxide must be added to the combustion bomb prior to its closure with the sample so that the Cl_2 is converted to Cl^- and enters the aqueous solution. Similarly, without a reducing agent, organic bromine compounds form about 80–85% bromine, Br_2 , and 15–20% HBr in aqueous solution. Aqueous arsenious oxide reduces the Br_2 to aqueous HBr during the oxidation reaction when the reducing agent is added to the bomb prior to closure. The combustion process for organic fluorine compounds give aqueous HF as the only fluorine combustion product while organic iodine compounds yield crystalline elemental iodine as the singular iodine-containing product. For additional information, the reader should examine 56ROS, 62SKI, and 79SUN/MAN.

The large differences between the experimental and estimated values for $\Delta_f H^\circ(\text{gas})$ and $\Delta_f H^\circ(\text{solid})$ for decafluorobiphenyl (322.8 and 337.84 $\text{kJ}\cdot\text{mol}^{-1}$, respectively) are not easily explained. The study reported by 79PRI/SAP2 for the combustion of $\text{C}_{12}\text{F}_{10}$ indicates that CO_2 , CF_4 , and F_2 are the only products of combustion in excess oxygen. Several reasons may explain the large differences. Possible explanations include: the energy corrections for the interactions between fluorine atoms in $\text{C}_{12}\text{F}_{10}$ may be different than those which are currently viewed; or, the quantitative determinations of the CF_4 and F_2 as combustion products may be in error.

We have attempted to correct for the interactions between halogen atoms in the various halogen families using *cis*-, *ortho*-, or *meta* corrections, but success here is limited.

2.6. Comparison with an Extended Second-order Group Additivity Scheme

An extended multi-parameterized second-order group-additivity estimation scheme has been developed by J.B. Pedley and co-workers (86PED/NAY).

The Pedley scheme is limited to the estimation of enthalpies of formation of organic compounds at 298.15 K in the gas phase. The additional parameterization accounts more comprehensively for nearest- and next-to-nearest neighbor interactions than the estimation scheme developed by S.W. Benson and co-workers and used in this work.

The details of the Pedley scheme are discussed in a cursory manner in Appendix 3 and more fully in 86PED/NAY. A comparison of estimated values from the Pedley scheme and that used in this work has been made for 20 hydrocarbons and 20 organic oxygen compounds. The results indicate that differences between literature and estimated values are about the same for the two groups of compounds tested.

2.7. Summary and Conclusions

We have demonstrated the successful extension of the second-order group-additivity method for the estimation of $\Delta_f H^\circ$, C_p° , and S° at 298.15 K to liquid and solid organic compounds. A re-examination of group values for the gas phase was performed in order to maintain internal consistency with the condensed phase. This work has been carried out for 1512 organic compounds containing the elements carbon, hydrogen, oxygen, nitrogen, sulfur, and halogens. A total of over 3700 comparisons between literature and estimated values have been made for $\Delta_f H^\circ$, C_p° , and S° in the gas, liquid, and solid phases. Overall, for the compounds covered, the estimation of $\Delta_f H^\circ$ showed that 67 percent of the residuals were $< \pm 4 \text{ kJ}\cdot\text{mol}^{-1}$, 16 percent were $> \pm 4$ but $< \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$, and 17 percent were $> \pm 8 \text{ kJ}\cdot\text{mol}^{-1}$. Values for C_p° showed that 80 percent of the residuals are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 10 percent are $> \pm 4$ but $< \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, and 10 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$. Values for S° show that 76 percent of the residuals are $< \pm 4 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, 14 percent are $> \pm 4$ but $< \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$, and 10 percent are $> \pm 8 \text{ J}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$.

The groups and group values developed in this work should be helpful to thermochemists and chemical engineers for the estimation of enthalpies of formation, heat capacities, and entropies at 298.15 K and 101,325 Pa when their needs for predicted values of these thermodynamic properties arise. This estimation technique can also be used to establish whether a new experimentally determined value for $\Delta_f H^\circ$, C_p° , or S° comes within the range of expectations of group additivity as dictated by the experience already shown with this method.

Comparisons in Appendix 2 between literature values for the enthalpy and entropy of fusion and the enthalpy of vaporization, corrected from either T_m or T_b to 298.15 K, with corresponding differences [$\Delta_f H^\circ(\text{solid}) - \Delta_f H^\circ(\text{liq})$], [$S^\circ(\text{solid}) - S^\circ(\text{liq})$], and [$\Delta_f H^\circ(\text{liq}) - \Delta_f H^\circ(\text{g})$], respectively, show that internal consistency does exist for the compounds tested.

The limited comparison of hydrocarbon and oxygen-containing compounds in Appendix 3 suggest that the extra effort taken in the Pedley scheme to account for nearest- and next-to-nearest neighbor interactions may have either a very small or even negligible effect upon reducing the degree of differences between literature and estimated values for enthalpies of formation at 298.15 K in the gas phase.

3. Tables of C-H-N-O-S-Halogen Compounds

TABLE 1. Summary of tables of C-H-N-O-S Halogen Families

Table 1.	C-H-N-O-S-Halogen Families
Table 2.	Listing of Groups and Group Values
Table 3.	General Definitions and Examples of Notations for Molecular Groups

Hydrocarbon Compounds

Table	Name	Description	No. of Compounds
Table 4.	<i>n</i> -Alkanes	normal alkanes	25
Table 5.	<i>t</i> -alkanes	tertiary branched alkanes	35
Table 6.	<i>q</i> -Alkanes	quaternary branched alkanes	16
Table 7.	<i>n</i> -Alkenes	linear alkenes	32
Table 8.	<i>s</i> -Alkenes	branched alkenes	34
Table 9.	Alkynes	alkyne hydrocarbons	28
Table 10.	Aromatic CH-01	aromatic hydrocarbons	42
Table 11.	Aromatic CH-02	aromatic hydrocarbons	80
Table 12.	Cyclic CH-01	cyclic hydrocarbons	40
Table 13.	Cyclic CH-02	cyclic hydrocarbons	48
Table 14.	Cyclic CH-03	cyclic hydrocarbons	47
Total Hydrocarbon compounds			427

CHO Compounds

Table	Name	Description	No. of Compounds
Table 15.	Alcohols	alcohols, diols, triols, phenols	69
Table 16.	Ethers	linear, branched, and cyclic ethers	52
Table 17.	Aldehydes	aldehydes	16
Table 18.	Ketones	ketones	42
Table 19.	Acids	linear, branched, cyclic, and aromatic acids	89
Table 20.	Anhydrides	anhydrides	11
Table 21.	Esters	esters and lactones	74
Table 22.	Peroxides	peroxides	7
Table 23.	Hydroperoxides	hydroperoxides	9
Table 24.	Peroxyacids	peroxyacids	8
Table 25.	Carbonates	carbonates	3
Total CHO compounds			381

CHN Compounds

Table	Name	Description	No. of Compounds
Table 26.	Amines	Linear, branched, cyclic, aromatic	50
Table 27.	Imines	imines	2
Table 28.	Nitriles	linear, branched, cyclic, aromatic	27
Table 29.	Hydrazines	hydrazines	6
Table 30.	Diazenes	diazenes	14
Table 31.	Azides	azides	6
Table 32.	Cyclic CHN	heterocyclic nitrogen compounds	32
Total CHN compounds			137

TABLE 1. C-H-N-O-S-Halogen families (Continued)

CHNO Compounds			
Table	Name	Description	No. of compounds
Table 33.	Amides	Linear, Branched, Cyclic, Aromatic	28
Table 34.	Ureas	Ureas	24
Table 35.	Amino acids	Amino acids and peptides	38
Table 36.	Nitroso compounds	Nitroso and cyanato compounds	9
Table 37.	Nitro compounds	Linear, branched, cyclic, aromatic	50
Table 38.	Nitrites	Nitrites	3
Table 39.	Nitrates	Nitrates	6
Table 40.	Nitramines	Nitramines	10
Table 41.	Cyclic CHNO	Cyclic amides	3
Total CHNO compounds			171
CHS and CHSO Compounds			
Table	Name	Description	No. of compounds
Table 42.	Thiols	Linear, branched, cyclic, aromatic	31
Table 43.	Sulfides	Linear, branched, aromatic	33
Table 44.	Disulfides	Disulfides	8
Table 45.	Sulfoxides	Sulfoxides	6
Table 46.	Sulfones	Linear, branched, aromatic	38
Table 47.	Sulfites	Sulfites	5
Table 48.	Sulfates	Sulfates	4
Table 49.	Cyclic CHS	Heterocyclic sulfur compounds	13
Total CHS and CHSO compounds			138
Halogen Compounds			
Table	Name	Description	No. of compounds
Table 50.	Fluorides	CHF and CHFO compounds	46
Table 51.	Chlorides	CHCl and CHClO compounds	116
Table 52.	Bromides	CHBr and CHBrO compounds	39
Table 53.	Iodides	CHI and CHIO compounds	39
Table 54.	Mixed Halogen compounds	CHClF, CHClBr, CHBrF, CHFI compounds	18
Total Halogen compounds			258
Total of all compounds			1512

TABLE 2. Listing of groups and group values

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CH Groups									
C-(H) ₃ (C)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (C) ₂	-20.63	22.89	39.16	-25.73	30.42	32.38	-29.41	21.92	23.01
C-(H)(C) ₃	-1.17	20.08	-53.60	-4.77	21.38	-23.89	-5.98	-48.81	-16.89
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₄	19.20	16.53	-149.49	17.99	10.24	-98.65	12.47	-83.63	-33.19
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
-CH ₃ corr (tert/quat)	-1.80	0.00	0.00	-1.77	0.00	0.00	-2.70	0.00	0.00
-CH ₃ corr (quat/quat)	-0.64	0.00	0.00	-0.64	0.00	0.00	-2.24	0.00	0.00
C _d -(H) ₂	26.32	21.38	115.52	21.75	28.37	86.19	22.43		
C _d -(H)(C)	36.32	18.74	33.05	31.05	24.60	28.58	25.48		
C _d -(C) ₂	44.14	15.10	-50.84	39.16	23.22	-29.83	32.97		
C _d -(H)(C _d)	28.28	18.54	27.74	22.18	31.67	13.30	17.53	35.65	21.75
C _d -(C)(C _d)	36.78	17.57	-61.33	30.42	26.19	-41.92	27.91		
C _d -(C _d)(C _B)							56.07		
C _d -(H)(C _B)	28.28	18.54	27.74	22.18	31.67	13.30	17.53	35.65	21.75
C _d -(C)(C _B)	37.95	15.90	-51.97	38.58					
C _d -(H)(C _i)	28.28	18.54	27.74	22.18	31.67	13.30	17.53	35.65	21.75
C-(H) ₄ , Methane	-74.48	35.73	206.92						
C _d -(C _B) ₂	32.88			30.83	25.10		49.91	32.50	
C-(H) ₂ (C)(C _d)	-20.88	20.63	38.20	-25.73	29.29	31.67	-24.35		
C-(H)(C) ₂ (C _d)	-1.63	27.49	-50.38	-5.02	30.12	-28.07	-6.49		
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (C _d)	22.13	9.16	-150.23	20.79	28.74	-108.20	12.51		
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
C-(H)(C)(C _d) ₂	-1.17	20.08	-53.60	-4.77	21.38	-23.89	-5.98	-48.81	-16.89
C-(H) ₂ (C _d) ₂	-18.92	24.77	42.08	-24.43	40.88	19.32	-21.60		
C-(H) ₂ (C _d)(C _B)				-24.73					
C-(H)(C)(C _d)(C _B)				-6.90					
cis (unsat) corr	4.85	-8.03	5.06	5.27	0.00	0.00	5.73	0.00	0.00
tert-Butyl cis corr	17.24	0.00	0.00	17.48	0.00	0.00	17.57	0.00	0.00
C _r -(H)	113.50	22.55	101.96	104.47	39.96	67.57	110.34		
C _r -(C)	115.10	13.22	26.32	107.15	25.59	14.25	101.66		
C _r -(C _d)	121.42	10.71	39.92	114.77					
C _r -(C _B)	120.76	10.17	17.77	119.00			103.28	32.30	
C _r -(C _i)	120.76	14.27	25.94	104.80			103.28		
C-(H) ₃ (C)(C _i)	-19.70	20.97	42.80	-22.13	30.39	32.36	-29.41		
C-(H)(C) ₂ (C _i)	-3.16	17.45	-45.69						
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (C _i)				22.83			26.38		
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
C-(H) ₂ (C _i) ₂	-41.14			-39.08					
C-(C) ₂ (C _i) ₂				20.67					
C _B	142.67	15.86	26.28	134.68	30.04	14.39	131.08		
C _B -(H)(C _B) ₂	13.81	13.61	48.31	8.16	22.68	28.87	6.53	20.13	22.75
C _B -(C)(C _B) ₂	23.64	9.75	-35.61	19.16	10.10	-19.50	13.90	-23.26	-5.50
C _B -(C _d)(C _B) ₂	24.17	14.12	-33.85	19.12	9.44	-9.04	20.27	-20.00	-10.00
C _B -(C _i)(C _B) ₂	24.17	14.12	-33.85	19.12	9.44	-9.04	20.07	-20.00	-10.00
C _B -(C _B) ₃	21.66	13.12	-36.57	17.21	17.07		17.03	-1.72	-6.00
C-(C) ₂ (C _B) ₂							52.81		
C-(H) ₂ (C)(C _B)	-21.34	25.61	42.59	-24.81	22.90	47.40	-22.10	49.38	26.90
C-(H)(C) ₂ (C _B)	-4.52	22.45	-48.00	-5.82	17.50	-13.90	-3.50		
C-(C _B)(C) ₃	18.28	18.28	-147.19	18.70	5.17	-96.10	21.57		
C-(H) ₂ (C _B) ₂	-46.43			-26.50	32.91	51.97	-21.44	69.06	22.85
C-(H)(C)(C _B) ₂				-21.47	11.50	28.12	16.40	43.55	
C-(H)(C _B) ₃	-6.86						34.48	63.64	-12.62
C-(C _B) ₃ (C)							116.25	39.83	

TABLE 2. Listing of groups and group values

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CH Groups									
C-(C _B) ₄	27.04						64.89	58.74	
C _{BF} -(C _{BF})(C _B) ₂	20.10	0.00	0.00	15.83	9.52	-5.54	14.10	2.30	-6.00
C _{BF} -(C _B)(C _{BF}) ₂	16.00			11.50			12.00	5.77	2.00
C _{BF} -(C _{BF}) ₃	3.59			-0.90			1.94	8.00	7.00
C _B -(C _B) ₂ (C _{BF})							-8.77		
C _B -(C _B)(C _{BF}) ₂	22.46						47.93		
<i>ortho</i> corr, hydrocarbons	1.26	6.40	-2.50	3.26	3.50	0.00	5.00	0.00	0.00
<i>meta</i> corr, hydrocarbons	-0.63	0.71	0.00	0.00	0.00	0.00	2.00	0.00	0.00
Cyclopropane rsc (unsub)	115.15	-12.73	134.86	111.58	-28.53				
Cyclobutane rsc	110.89	-19.34	126.04	106.64	-10.68	51.48	114.43		
Cyclopentane rsc (unsub)	26.75	-31.44	116.22	22.84	-23.32	42.24	34.00		
Cyclohexane rsc (unsub)	0.68	-31.07	78.18	-1.77	-26.21	10.07	10.94		
Cycloheptane rsc	26.34	-37.14	73.97	23.50	-32.19	15.89			
Cyclooctane rsc	40.65	-43.17	70.78	38.10	-27.88	2.96			
Cyclononane rsc	52.91			50.40					
Cyclodecane rsc	51.99			50.61					
Cycloundecane rsc	47.56			47.55					
Cyclododecane rsc	17.31						46.27		
Cyclotridecane rsc	21.84			24.83					
Cyclotetradecane rsc	49.37						37.48		
Cyclopentadecane rsc	8.03						65.09		
Cyclohexadecane rsc	8.41						67.14		
Cycloheptadecane rsc	-13.59						69.56		
Cyclopropene rsc	223.26								
Cyclobutene rsc	125.81	-11.67	126.77						
Cyclopentene rsc (unsub)	24.18	-26.53	113.76	21.45	-15.82	48.37			
Cyclohexene rsc	5.61	-19.50	95.69	2.04	-20.26	29.34			
Cycloheptene rsc	21.81								
Cyclooctene rsc	24.65			18.26					
1,3-Cyclopentadiene rsc	24.07			23.95					
1,3-Cyclohexadiene rsc	17.14			16.41	-26.56	50.18			
1,4-Cyclohexadiene rsc	-2.69			-5.64	-34.22	36.41			
1,3-Cycloheptadiene rsc	27.54								
1,5-Cyclooctadiene rsc	39.34			36.42	-7.45	23.35			
1,3,5-Cycloheptatriene rsc	16.84	-18.63	102.26	18.59	-54.00	84.96			
Cyclooctatetraene rsc	71.37	-26.31	116.38	77.07	-68.18	113.89			
Spiropentane rsc	248.50	-19.97	286.59	242.58	2.60	162.81			
Cyclopropane (sub) rsc	105.95			96.58					
Cyclopentane (sub) rsc	19.55	-27.87	118.39	23.59	-23.32	56.65	34.00		
Cyclohexane (sub) rsc	-0.39	-22.82	83.97	-2.06	-26.21	25.10	10.30		
Cyclopentene (sub) rsc	24.31	-24.50	117.11	19.82	-15.82	48.37			
Naphthalene (unsub)	0.00	11.83	-19.66	0.00	0.00	0.00	0.00	0.00	0.00
Naphthalene (1 sub)	0.00	14.39	-21.50	0.00	0.00	0.00	0.00	0.00	0.00
Naphthalene (2 sub)	0.00	16.48	-23.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>cis</i> -Decalin rsc	127.00	-183.44		-4.02	-54.12	53.75			
<i>trans</i> -Decalin rsc	123.81	-183.24		-15.22	-57.63	53.67			
<i>cis</i> -Hexahydroindan rsc	19.51			16.39	-41.52	86.59			
<i>trans</i> -Hexahydroindan rsc	15.16			13.29	-46.00	79.98			
2,2-Metacyclopentane rsc	52.08						55.06	-24.92	
2,2-Metaparcyclopentane rsc	99.35						109.46	-4.02	
2,2-Paracyclopentane rsc	125.09						127.26	-13.18	-1.92
3,3-Paracyclopentane rsc	50.95						65.53	14.90	
Adamantane rsc	-6.14						3.18		
Fluoranthene rsc	63.21	62.47	59.66	-4.43	-5.92				

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CH Groups									
Bicyclo[2.2.2]octane rsc	27.12				-67.59	-63.45	41.52		
Bicyclo[3.3.3]undecane rsc	99.06						124.10		
cis-Bicyclo[6.1.0]nonane rsc	115.55			109.35					
Bicyclo[1.1.0]butane rsc	260.70			254.70					
Bicyclo[3.1.0]hexane rsc	123.16			117.56					
Bicyclo[2.2.1]hepta-2,5-diene rsc	125.29			124.87					
Tetracyclo-[3.2.02,7.04,6]heptane rsc	366.75			356.45					
Tricyclo[2.2.1.02,6]heptane rsc	148.67			139.67					
Bicyclo[2.2.1]hept-2-ene rsc	82.79			73.58			102.73		
Bicyclo[2.2.1]heptane rsc	43.49			45.39			57.01		
Bicyclo[4.1.0]heptane rsc	106.99			101.39					
Pentacyclo-[4.2.0.02,5.03,8.04,7]-octane rsc	674.60						632.84		
Bicyclo[2.2.2]oct-2-ene rsc	33.64						56.36		
Bicyclo[4.2.0]octane rsc	100.72			95.72					
Bicyclo[5.1.0]octane rsc	109.42			103.62					
trans-Bicyclo[6.1.0]nonane rsc	107.05			107.25					
Bicyclo[3.3.1]nonane rsc	19.25						39.63		
cis-Bicyclo[3.3.0]octane rsc	33.22			27.92					
trans-Bicyclo[3.3.0]octane rsc	59.52			54.72					
CHO Groups									
CO-(H) ₂ , Formaldehyde	-108.60	35.40	224.54						
CO-(C)(CO)	-121.29			-135.04			-140.75		
CO-(H)(CO)	-105.98								
CO-(CO)(C _B)	-112.30						-117.75		
CO-(O)(CO)	-123.75			-123.30	40.63		-120.81		
CO-(C _d)(O)	-136.73	24.56	62.59	-155.56	48.16		-134.10	43.75	32.90
CO-(C)(O)	-137.24	24.56	62.59	-149.37	44.98	32.72	-153.60	44.98	32.13
CO-(H)(O)	-124.39	29.00	147.03	-142.42	65.10	94.68			
CO-(O) ₂	-111.88			-122.00	31.46		-123.00	4.25	-42.92
CO-(H)(C _d)	-126.96			-153.05					
CO-(C _B) ₂	-110.00			-119.00			-116.00	109.33	
CO-(C)(C _B)	-148.82			-145.22	73.35		-143.70	71.38	23.72
CO-(H)(C _B)	-121.35			-138.12	54.22		-160.18		
CO-(O)(C _B)	-125.00			-140.00	48.16		-145.00	43.75	32.13
CO-(C) ₂	-132.67	23.43	64.31	-152.76	52.97	33.81	-157.95		
CO-(H)(C)	-124.39	29.00	147.03	-142.42	65.10	93.55			
CO-(C)(C _d)					27.07				
O-(CO) ₂ , aliphatic	-214.50	-1.08	34.16	-230.50	5.28		-235.00		
O-(CO) ₂ , aromatic	-238.30			-220.90			-207.00		
O-(C _d)(CO)	-198.03			-201.42	19.58				
O-(C)(CO)	-188.87	11.80	36.03	-196.02	19.58	38.28	-210.60	-6.00	12.09
O-(H)(CO)	-254.30	16.23	101.71	-285.64	37.82	38.28	-282.15	44.60	21.78
O-(C _B)(CO)	-167.00			-165.50			-170.00	29.08	45.32
O-(C)(O)	-20.75			-23.50			-30.20		
O-(H)(O)	-72.26			-101.75			-105.30		
O-(C _d) ₂	-139.29			-137.32					
O-(H)(C _d)					37.78				
O-(C)(C _d)	-129.33			-133.72	51.21				
O-(C _B) ₂	-77.66			-85.27		23.31	-96.20	15.90	3.14
O-(C)(C _B)	-92.55			-104.85	8.10		-122.87		
O-(H)(C _B)	-160.30	18.16	121.50	-191.75	44.64	43.89	-199.25	29.25	28.62
O-(C) ₂	-101.42	18.54	29.33	-110.83	24.27	26.78	-119.00		
O-(H)(C)	-159.33	18.16	121.50	-191.50	44.64	43.89	-199.66	29.25	28.62
C _d -(H)(CO)	32.30	15.61	35.19	26.61	28.12		7.82	-18.66	27.53
C _d -(C)(CO)					18.62				

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHO Groups									
C _d -(O)(C _d)	36.78	17.57	-61.34	30.42	26.19	-41.92	27.91		
C _d -(O)(C)	44.14	15.10	-50.84	39.08	23.22	-29.83	32.97		
C _d -(O)(H)	36.32	18.74	33.05	31.05	24.60	28.58	25.48		
C _r -(CO)							144.52		
C _B -(CO)(C _B) ₂	15.50			10.50	4.39		8.15	-42.89	0.08
C _B -(O)(C _B) ₂	-4.75	15.86	-43.72	-5.61	39.71	-10.59	1.00	-0.29	1.59
C-(H) ₂ (CO) ₂	-30.74			-23.06	15.56		-19.10		
C-(CO)(C) ₃	23.93			26.15	7.99	-85.98	24.02	-114.10	
C-(H)(CO)(C) ₂	-0.25			-3.89	17.41	-24.52	-9.83	-80.51	
C-(H) ₂ (CO)(C)	-21.84	24.69	39.58	-24.14	29.29	39.87	-27.90	21.92	24.73
C-(H) ₃ (CO)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (CO)(C _d)	-16.95			-19.62					
C-(H) ₂ (CO)(C _i)	-25.48			-26.61					
C-(H) ₃ (CO)(C _B)	-16.20			-11.67					
C-(H)(CO)(C)(C _B)							14.81		
C-(H)(O)(CO)(C)	126.63			123.43	7.44	-46.71	-14.39	-58.45	8.08
C-(O) ₄	-152.46			-133.34					
C-(H)(O) ₃	-113.97			-107.74	21.71				
C-(O) ₃ (C)	-114.39			-99.54					
C-(O) ₂ (C) ₂	-53.56			-41.30					
C-(H)(O) ₂ (C)	-57.78			-51.42	12.38				
C-(H) ₂ (O) ₂	-62.22			-62.89	39.92	23.85			
C-(H) ₂ (O)(C _B)	-33.76			-29.17	46.48				
C-(H) ₂ (O)(C _d)	-27.49	17.74	37.49	-28.62	41.30				
C-(H)(CO)(C)(C _B)							-14.39		
C-(H)(CO)(C _B) ₂							3.72		
C-(O)(C _B) ₃							60.46	57.49	
C-(O)(C) ₃ (ethers,esters)	9.50	14.60	-141.92	0.79	20.46	-94.68	-0.50		
C-(H)(O)(C) ₂ (ethers,esters)	-19.46	17.78	-52.80	-21.00	25.56	-25.31	-20.08		
C-(O)(C) ₃ (alcohols,peroxides)	-13.50	15.73	-144.60	-11.13	65.58	-122.48	-12.25	-85.48	-14.77
C-(H)(O)(C) ₂ (alcohols,peroxides)	-26.10	19.96	-43.05	-27.60	49.83	-29.83	-29.08	4.77	6.95
C-(H) ₂ (O)(C)	-32.90	20.33	43.43	-35.80	33.64	32.59	-33.00	21.92	24.73
C-(H) ₃ (O)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
O-(CO)(O)	-88.00			-90.00			-80.50		
C-(C) ₂ (O)(C _B)	15.30			25.80			29.30		
C-(H)(C)(O) ₂							-52.50		
Glutaric anhydride rsc	20.89						8.91		
Succinic anhydride rsc	4.76			-11.08			-10.60		
Phthalic anhydride rsc	30.66						-5.52		
Cyclopentanone rsc	22.85			15.10					
Cyclohexanone rsc	10.50	-31.82	66.98	5.60	-25.61	11.29			
Cycloheptanone rsc	10.76			6.31					
Cyclooctanone rsc	7.33			9.01			37.38		
Cyclononanone rsc	20.43			22.57			55.28		
Cyclodecanone rsc	15.70			17.73					
Cycloundecanone rsc	19.39			20.53					
Cyclododecanone rsc	12.91			18.02			47.11		
Cyclopentadecanone rsc	9.41						74.77		
Cycloheptadecanone rsc	4.87						89.49		
Cyclobutane-1,3-dione rsc	140.48						94.10		
Ethylene oxide rsc	114.62	-10.92	132.00	104.82	-23.90	80.50			
Trimethylene oxide rsc	107.35				-22.38				
Furan rsc	-12.18			-9.97					
Tetrahydrofuran rsc	24.28	-28.73	113.66	17.70	-28.49	47.18	14.60		
Tetrahydropyran rsc	5.71			1.32	-42.22		0.80		

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHO Groups									
1,3-Dioxolane rsc	29.06			18.75	-37.74				
1,3-Dioxane rsc	10.90			4.40	-42.26				
1,4-Dioxane rsc	19.15	-24.34	73.16	9.76	-29.50	86.28	-12.00		
1,3-Dioxepane rsc	25.52			20.01	-49.20				
Trioxane rsc	25.02								
Tetraoxane rsc	34.23								
β -Propiolactone rsc	97.95			75.43	-5.40	31.85			
γ -Butyrolactone rsc	34.98			10.16	-16.61	21.56			
γ -Valerolactone rsc	26.06			4.75					
δ -Valerolactone rsc	42.51			19.19	-16.74	10.77			
Caprolactone rsc					-21.92	-4.92			
Undecanolactone rsc					-28.12	-33.05			
Ethylene carbonate rsc							23.90		
Cyclobutane methyl carboxylate rsc	75.21			79.08					
Bicyclobutane methyl carboxylate rsc	222.27			219.98					
1,4-Dimethylcubane dicarboxylate rsc	595.80						590.73		
2-Deoxy-D-ribose rsc							0.25		
β -D-Ribose rsc							12.65		
α -D-Glucose rsc							6.30		
COOH-COOH (<i>ortho</i> corr)							34.14	15.00	8.96
COOH-COOH (<i>meta</i> corr)	-23.94						13.14	30.00	0.00
CH ₃ O-COOH (<i>ortho</i> corr)	15.00						23.00		
CH ₃ O-COOH (<i>meta</i> corr)	5.00						5.00		
OH-OH (<i>ortho</i> corr)	7.00						16.00		
OH-OH (<i>meta</i> corr)	0.00						2.00		
OH-COOH (<i>ortho</i> corr)	-20.00						0.00		
CHN and CHNO Groups									
C-(H) ₃ (N)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (C)(N)	-28.30	22.68	42.26	-30.80	30.42	32.38	-34.00	21.92	23.01
C-(H)(C) ₂ (N)	-16.70	18.62	-63.55	-14.65	28.28	-20.00	-13.90		
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (N)	0.29	18.41	-152.59	5.10	19.66	-87.99	1.00	-84.14	
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
C-(H) ₂ (N) ₂	-30.00						-26.00		
C-(H) ₂ (C _B)(N)	-24.14			-26.09	19.79		-33.31		
N-(H) ₂ (C) (first, amino acids)	19.25	24.35	124.40	0.33	62.59	71.71	-6.30	32.00	39.00
N-(H) ₂ (C) (second, amino acids)	19.25	24.35	126.90	0.33	62.59	71.71	-46.00	71.27	48.75
N-(H)(C) ₂	67.55	12.28	33.96	51.50	59.37	32.09	47.80	-8.00	
N-(C) ₃	116.50	15.10	-61.71	112.00	26.11	-38.62	101.00	-39.00	
N-(H) ₂ (N)	47.70	26.36	122.18	25.30	49.41	60.58	18.97		
N-(H)(C)(N)	89.16			75.00	49.04	22.05			
N-(C) ₂ (N)	120.71			119.00	41.67	-26.94			
N-(C _B) ₂ (N)							137.35		
N-(H)(C _B)(N)	87.50			73.40			66.90		
N-(CO) ₂ (N)							73.62		
N-(H)(C _d) ₂	83.55			50.50			45.40		
N-(C)(C _d) ₂	120.64			97.38			88.92		
N-(H) ₂ (C _B)	19.25	24.35	126.90	-11.00	62.59	71.71	-21.60	26.00	70.00
N-(H)(C)(C _B)	59.00			26.25	65.20		36.55	-50.00	
N-(C) ₂ (C _B)	126.40			109.40	10.75		96.50	-36.50	
N-(C)(C _B) ₂	120.44			97.38	7.95		89.30		
N-(H)(C _B) ₂	83.55			50.50			45.40	-3.00	

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHN and CHNO Groups									
N-(C _B) ₃	123.15			121.80			107.50	-39.00	
N _T -(C)	81.46			73.68					
N _T -(C _B)	69.00	10.07	47.01	54.50	19.75	36.40	57.00		
N _A -(C)	109.50			104.85			103.00		
N _A -(C _B)	109.50			104.85			103.00		
N _A -(oxide)(C)	40.80			22.65					
C-(H) ₂ (C)(N _A)	-20.70			-25.70			-29.41		
C-(H)(C) ₂ (N _A)	-2.66			-5.42					
C-(C) ₃ (N _A)	11.50			15.50			10.50		
C _d -(H)(N)	-16.00			-15.50			-13.00		
C _d -(C)(N)	-5.74			-5.62			-3.95		
C _B -(N)(C _B) ₂	-1.30	16.07	-43.53	1.50	15.02	-24.43	9.75	13.00	-37.57
C _B -(NO)(C _B) ₂	21.50						23.00		
C _B -(NO ₂)(C _B) ₂	-1.45			-28.30	73.30	79.95	-32.50	50.96	110.46
C _B -(CNO)(C _B) ₂	-177.63						155.69		
C _B -(CN)(C _B) ₂	151.00	41.09	85.25	122.38	51.80	64.75	121.20		50.45
C _B -(N _A)(C _B) ₂	22.55			20.08			18.65		
C _B -(H)(N _i) ₂	6.30						0.25		
CO-(H)(N)	-124.39	29.00	147.03	-188.00	65.10	93.55			
CO-(C)(N)	-133.26	22.50	56.70	-185.00	49.16		-194.60	39.00	40.00
CO-(C _B)(N) (amides)							-177.75	111.50	
CO-(C _B)(N) (amino acids)							-177.75	37.00	
CO-(C _d)(N)	-171.80								
CO-(N) ₂	-111.00	32.40	96.00	-190.50			-203.10	124.00	69.00
N-(H) ₂ (CO) (amides, ureas)	-63.00	17.00	88.25	-63.90	43.01		-65.25	-15.50	18.00
N-(H) ₂ (CO) (amino acids)	-63.00			-63.90	43.01		-59.75	45.88	33.03
N-(H)(C)(CO) (amides, ureas)	-16.28			-17.10	23.51		-9.80	-36.00	
N-(H)(C)(CO) (amino acids)	-16.28			-17.10	23.51		5.50	3.30	
N-(C) ₂ (CO)	45.00			62.00	13.93		55.00		
N-(H)(C _B)(CO)	-20.84						-3.50	-41.00	
N-(H)(CO) ₂	-91.00						-30.80	-157.02	
N-(C)(CO) ₂	-11.64			56.20			64.00		
N-(C _B)(CO) ₂	9.12								
N-(C _B) ₂ (CO)							60.85		
N-(C)(C _B)(CO)							72.00		
C-(H) ₃ (CN), Acetonitrile	74.04	52.22	252.60	40.56	91.46	149.62			
C-(H) ₂ (C)(CN)	94.52	47.86	167.25	66.07	83.01	106.02	69.85	72.80	96.15
C-(H)(C) ₂ (CN)	113.50	44.94	67.86	81.50	83.09		69.00		
C-(C) ₃ (CN)	137.96			116.20	69.91	-17.91	102.07		
C-(C) ₂ (CN) ₂								44.60	74.57
C-(H) ₂ (C _d)(CN)	95.31			66.40					
C _d -(H)(CN)	146.65	42.38	158.41	117.28	80.42	92.72			
C _T -(CN)	264.60			250.20					
C-(H) ₃ (NO ₂), Nitromethane	-74.86	57.32	284.14	-112.60	105.98	171.75			
C-(H) ₂ (NO ₂) ₂ , Dinitromethane	-58.90			-104.90					
C-(H)(NO ₂) ₃ , Trinitromethane	-0.30			-32.80			-48.00		
C-(NO ₂) ₄ , Tetranitromethane	82.30			38.30					
C-(H) ₂ (C)(NO ₂)	-60.50	53.14	203.60	-93.50	97.74		-99.00		
C-(H)(C) ₂ (NO ₂)	-53.00	49.58	115.32	-82.50			-89.00		
C-(C) ₃ (NO ₂)	-36.65			-61.20			-76.55		
C-(H) ₂ (C _B)(NO ₂)	-62.00			-82.76			-81.00		
C-(H)(C)(NO ₂) ₂	-36.80			-88.80			-91.50		
C-(C) ₂ (NO ₂) ₂	-28.50			-77.20			-90.30	71.38	
C-(H)(C)(CO)(N)	-18.70						-11.65	-22.85	-4.00
C-(H) ₂ (CO)(N)	-3.10						-30.95	21.92	24.00
C-(H)(C _B)(CO)(N)								61.21	

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHN and CHNO Groups									
O-(C)(NO)	-24.23	37.49	166.11	-46.50					
O-(C)(NO ₂)	-79.71	51.46	191.92	-108.96	96.40	127.50	-124.00		
N-(H)(C)(NO ₂)							-16.50	65.73	
N-(H)(C _B)(NO ₂)								-47.53	
N-(H)(CO)(NO ₂)							-14.00		
N-(C)(NO ₂) ₂	100.30			53.50					
N-(C)(C _B)(NO ₂)	183.00			167.00			150.50		
N-(C) ₂ (NO)	90.00			59.00			55.00		
N-(C) ₂ (NO ₂)	88.00			50.00			40.00		
C-(H) ₂ (C)(N ₃)				321.70					
C-(H)(C) ₂ (N ₃)	274.00			255.00					
C-(H) ₂ (C _B)(N ₃)	347.00			327.40					
C-(C _B) ₃ (N ₃)	328.60						346.50		
C _B -(N ₃)(C _B) ₂	320.00			303.50					
Zwitterion energy, aliphatic	0.00	0.00	0.00	0.00	0.00	0.00	-55.10	-44.50	-13.40
Zwitterion energy, aromatic I	0.00	0.00	0.00	0.00	0.00	0.00	-32.00	-20.50	-13.00
Zwitterion energy, aromatic II	0.00	0.00	0.00	0.00	0.00	0.00	-11.00	5.00	-9.00
Ethyleneimine rsc	115.53	-5.13	137.90	101.98					
Pyrrolidine rsc	26.71	-22.29	118.45	20.36	-24.48	42.40			
Piperidine rsc	3.14			-1.09	-29.79	15.98			
Hexamethyleneimine rsc					-36.86				
Octahydroazocine rsc					-42.31				
Pyrrolizidine rsc	35.42			18.87					
3,5-Dimethylpyrrolizidine rsc	38.46			20.05					
Trimethyl cyanurate rsc	-95.00						-120.40		
Succinimide rsc	25.70						16.70		
Glutarimide rsc	28.23						17.57		
Azetidine rsc	116.00			102.00					
Pyrrole rsc	-30.48			-20.03			-17.84		
Cyclotetramethylenediazene rsc	12.86			-4.34					
Cyclotrimethylenediazene rsc	-10.47						-23.97		
Cyclopropanenitrile rsc	110.56			110.76	-28.53				
Cyclobutanenitrile rsc	91.39			98.69	-28.35				
Cyclopentanenitrile rsc	10.82			22.12	-37.27				
Cyclohexanenitrile rsc	-5.55			-0.05	-57.29				
N-Nitrosopiperidine rsc	45.20			48.70					
N-Nitropiperidine rsc	-13.91			-4.11			8.48		
R-salt rsc							195.30		
RDX rsc	32.00						30.00		
HMX rsc	17.00						32.00		
DINO-PMTA rsc							46.70		
cis-Azobenzene corr	48.40						49.10		
Azidocyclopentane rsc	29.42			27.02					
Azidocyclohexane rsc	-16.45			-17.95					
NO ₂ -NO ₂ (ortho corr)	44.00			45.25			40.60	3.76	
NO ₂ -NO ₂ (meta corr)	11.00			13.50			13.50	5.84	
NO ₂ -CH ₃ (ortho corr)	2.00			2.00			4.00		
NO ₂ -CH ₃ (meta corr)				-4.00					
NO ₂ -OH (ortho corr)	10.00			16.00			13.00		
NO ₂ -OH (meta corr)	6.00						0.00		
NO ₂ -NO ₂ (aliphatic-adjacent corr)	20.00			20.00			20.00		
NO ₂ -COOH (ortho corr)	25.00			30.00			25.00	0.00	
NO ₂ -COOH (meta corr)	14.00			16.00			14.00	0.00	

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHN and CHNO Groups									
NO ₂ -OH(<i>ortho</i> corr)	10.00			16.00			13.00		
NO ₂ -OH(<i>meta</i> corr)	6.00			0.00			0.00		
NH ₂ -NO ₂ (<i>ortho</i> corr)	-4.00			-4.00			-4.00	0.00	
NH ₂ -NO ₂ (<i>meta</i> corr)	-10.00			-10.00			-10.00		
(ONO ₂)-(ONO ₂)(aliphatic-adjacent corr)	15.10			15.90			16.00		
N _T -(CH ₃) (<i>ortho</i> corr)	-6.30			-4.00					
N _T -N _T (<i>ortho</i> corr)	85.06			83.16					
CH ₃ -CN (<i>cis</i> , unsat corr)	-6.00			-6.00					
NH ₂ -NH ₂ (<i>ortho</i> corr)							-3.00		
NH ₂ -NH ₂ (<i>meta</i> corr)							-10.00		
NH ₂ -COOH (<i>ortho</i> corr)				12.00			14.00	-4.71	
NH ₂ -COOH (<i>meta</i> corr)				2.00			4.00	-7.22	
CHS and CHSO Groups									
C-(H) ₃ (S)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (C)(S)	-23.17	20.90	41.87	-26.77	24.18	41.09			
C-(H)(C) ₂ (S)	-5.88	20.29	-47.36	-6.07	17.78	-16.61			
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (S)	13.52	17.02	-145.38	16.69	8.88	-86.86			
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
-CH ₃ corr (tert/quat)	-1.80	0.00	0.00	-1.77	0.00	0.00	-2.70	0.00	0.00
-CH ₃ corr (quat/quat)	-0.64	0.00	0.00	-0.64	0.00	0.00	-2.24	0.00	0.00
C-(H) ₂ (C _B)(S)	-18.53			-23.82					
C-(H) ₂ (C _d)(S)	-25.93			-32.44					
C-(H) ₂ (S) ₂	-25.10								
C _B -(S)(C _B) ₂	-4.75	15.86	43.72	-5.61	39.71	-10.59	1.00	-0.29	1.59
C _d -(H)(S)	36.32	18.74	33.05	31.05	24.60	28.58	25.48		
C _d -(C)(S)	45.73	14.64	-51.92						
S-(C)(H)	18.64	25.76	137.67	0.06	51.34	85.95			
S-(C _B)(H)	48.10	20.98	57.34	28.51	20.11	89.04			
S-(C) ₂	46.99	22.64	55.19	29.82	45.15	29.80			
S-(H)(C _d)	25.52								
S-(C)(C _d)	54.39								
S-(C _d) ₂	102.60	20.04	68.59						
S-(C _B)(C)	76.21			58.20	16.43	35.44	42.00		
S-(C)(S)	27.62	23.25	50.50	14.36	40.71	30.84			
S-(C _B)(S)	57.45						40.60		
S-(S) ₂	12.59	19.66	56.07						
S-(C _B) ₂	102.60	20.04	68.59	93.02	-35.10				
S-(H)(S)	7.95								
S-(H)(CO)	-5.90	31.92	130.54						
CO-(C)(S)	-132.67	23.43	64.31	-152.76	52.97	33.81		33.89	
C-(H) ₃ (SO)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (C)(SO)	-29.16			-36.88					
C-(H)(C) ₂ (SO)									
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (SO)	4.56			0.97					
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
C-(H) ₂ (C _d)(SO)	-27.56			-32.63					
<i>cis</i> correction	4.11	-8.03	5.06	5.27	0.00	0.00	5.73	0.00	0.00
C _B -(SO)(C _B) ₂	15.48			25.44	4.39		7.55	-42.89	0.08
O-(SO)(H)	-158.60								
O-(C)(SO)	-92.60								

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHS and CHSO Groups									
SO-(C) ₂	-66.78	37.15	75.73	-108.98	80.22	22.18			
SO-(C _B) ₂	-62.26								
SO-(O) ₂	-213.00								
SO-(C)(C _B)	-72.00								
C-(H) ₃ (SO ₂)	-42.26	25.73	127.32	-47.61	36.48	83.30	-46.74	67.45	56.69
C-(H) ₂ (C)(SO ₂)	-27.03			-33.76			-35.96		
C-(H)(C) ₂ (SO ₂)	-14.00								
-CH ₃ corr (tertiary)	-2.26	0.00	0.00	-2.18	0.00	0.00	-2.34	0.00	0.00
C-(C) ₃ (SO ₂)	1.52			2.00			3.78		
-CH ₃ corr (quaternary)	-4.56	0.00	0.00	-4.39	0.00	0.00	-4.35	0.00	0.00
-CH ₃ corr (quat/quat)	-0.64			-0.64			-2.24		
C-(H) ₂ (C _d)(SO ₂)	-29.49			-49.05					
C-(H)(C)(C _d)(SO ₂)	-71.99								
C-(H) ₂ (C _B)(SO ₂)	-29.80								
C-(H) ₂ (C _i)(SO ₂)	16.36								
C _B -(SO ₂)(C _B) ₂	15.48			25.44	4.39		7.55	-42.89	0.08
C _d -(H)(SO ₂)	51.58								
C _i -(C)(SO ₂)	64.01								
C _i -(SO ₂)	177.10								
SO ₂ -(C _d)(C _B)	-291.55								
SO ₂ -(C _d) ₂	-306.70								
SO ₂ -(C) ₂	-288.58	48.54	87.37	-341.14			-356.62	-9.55	32.10
SO ₂ -(C)(C _B)	-289.10								
SO ₂ -(C _B) ₂	-287.76						-305.40		
SO ₂ -(SO ₂)(C _B)	-325.18						-361.75		
SO ₂ -(O) ₂	-417.30								
SO ₂ -(C)(C _d)	-316.80								
SO ₂ -(C _i)(C _B)	-296.30								
O-(SO ₂)(H)	-158.60								
O-(C)(SO ₂)	-91.40								
Thiacyclopropane rsc	81.57	-10.76	122.10	75.32					
Thiacyclobutane rsc	80.98	-18.00	112.89	74.55	-10.54	40.57			
Thiacyclopentane rsc	6.41	-19.34	97.87	2.08	-14.19	31.08			
Thiacyclohexane rsc	-2.02	-24.91	66.85	-5.09	-21.47	9.12			
Thiacycloheptane rsc	20.53	-31.40	66.35	13.84					
2,5-Dihydrothiophene rsc	19.13			19.96					
Thiophene rsc	-43.54	-1.59	22.79						
2,3-Dihydrothiophene rsc	7.72								
CHX and CHXO Groups									
C-(H) ₃ (F), Methyl fluoride	-247.00	37.49	231.93						
C-(H) ₃ (Cl), Methyl chloride	-81.90	40.75	243.60						
C-(H) ₃ (Br), Methyl bromide	-37.66	42.43	254.94	-61.10					
C-(H) ₃ (I), Methyl iodide	14.30	44.14	263.14	-11.70	82.76				
C-(C)(F) ₃	-673.81	52.99	178.22	-709.07	73.18	135.56			
C-(H) ₂ (C)(F)	-221.12	33.66	146.80						
C-(H)(C) ₂ (F)	-204.46	30.55	55.76						
C-(C) ₃ (F)	-202.92								
C-(H)(C)(F) ₂	-454.74	42.22	164.32	-487.23	68.04				
C-(C) ₂ (F) ₂	-411.39	41.42	74.48	-400.37			-428.77		
C-(C)(Cl)(F) ₂	-462.70	57.32	169.45	-466.00	83.64	138.31			
C-(H)(C)(Cl)(F)	-271.14								
C-(C)(Cl) ₃	-81.98	68.18	202.14	-112.93	102.20	145.91			
C-(H)(C)(Cl) ₂	-79.10	50.69	183.28	-102.60	85.02	128.45			

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHX and CHXO Groups									
C-(H) ₂ (C)(Cl)	-69.45	37.53	159.24	-86.90	63.76	104.27	-85.65		
C-(C) ₂ (Cl) ₂	-79.56	54.40	95.41	-101.80	74.24				
C-(H)(C) ₂ (Cl)	-55.61	35.00	71.34	-71.17	66.02				
C-(C) ₃ (Cl)	-43.70	29.63	-24.26	-56.78					
C-(C)(Br) ₃		69.87	233.05						
C-(H)(C)(Br) ₂									
C-(H) ₂ (C)(Br)	-21.78	37.82	173.31	-42.65	66.00	113.00			
C-(C) ₂ (Br) ₂									
C-(H)(C) ₂ (Br)	-10.75	36.77	84.69	-27.31	59.24				
C-(C) ₃ (Br)	7.26	39.33	-13.46	-7.40					
C-(C)(I) ₃									
C-(H)(C)(I) ₂	108.78	51.04	228.45						
C-(H) ₂ (C)(I)	33.54	40.94	177.78	4.14	65.36		3.65		
C-(C) ₂ (I) ₂									
C-(H)(C) ₂ (I)	48.74	38.62	88.10	24.78					
C-(C) ₃ (I)	68.46	41.09	-3.21	48.60					
C-(H)(C)(Br)(Cl)	-18.45	51.88	191.21						
N-(C)(F) ₂	-32.64								
C-(H)(C)(Cl)(O)	-90.37	37.66	66.53						
C-(H) ₂ (I)(O)	15.90		170.29						
C-(C)(Cl) ₂ (F)	-322.54			-343.87	89.29	141.71			
C-(C)(Br)(F) ₂	-394.55				85.40	149.70			
C-(C)(Br) ₂ (F)									
C-(Br)(Cl)(F)									
C _d -(H)(F)	-165.12	28.45	137.24						
C _d -(H)(Cl)	4.37	32.75	147.85	-12.67	56.62				
C _d -(H)(Br)	50.94	34.10	159.91		79.13				
C _d -(H)(I)	102.36	36.82	169.45						
C _d -(C)(Cl)	-5.06		62.76	-2.23					
C _d -(F) ₂	-329.90	39.43	155.63						
C _d -(Cl) ₂	-11.51	46.86	175.41	-32.08	76.47	115.35			
C _d -(Br) ₂		51.46	199.16						
C _d -(I) ₂									
C _d -(Cl)(F)	-235.10	44.50	175.61						
C _d -(Br)(F)		45.19	177.82						
C _d -(Cl)(Br)		50.63	188.70						
C _r -(F)									
C _r -(Cl)		33.01	140.00						
C _r -(Br)		34.69	151.30						
C _r -(I)	35.53	158.41							
C _B -(F)(C _B) ₂	-181.26	26.10	67.52	-191.20	37.09	54.19	-194.00	32.05	39.79
C _B -(Cl)(C _B) ₂	-17.03	29.33	77.08	-32.20	35.27	55.47	-32.00	33.55	43.37
C _B -(Br)(C _B) ₂	36.35	29.65	88.60	19.90	40.91	74.85	13.50		54.45
C _B -(I)(C _B) ₂	94.50	32.70	98.26	73.70	45.17	61.08	70.40	40.08	
cis corr-(I)(I)	3.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C-(H) ₂ (CO)(Cl)	-44.26			-58.41			-74.75		
C-(H)(CO)(Cl) ₂	-40.40			-55.11					
CO-(C)(F)	-379.84			-419.59					
C-(C _B)(F) ₃	-691.79	52.30	179.08	-696.66					
C-(H) ₂ (C _B)(Br)	-29.49			-44.06					
C-(H) ₂ (C _B)(I)	7.31			-7.24					
C-(H) ₂ (C _B)(Cl)	-73.79			-92.56					
CO-(C)(Cl)	-200.54	42.09	176.66	-225.29	80.67				
CO-(C _B)(Cl)				-216.67	69.21		-212.99		
CO-(C)(Br)	-148.54			-175.49					
CO-(C)(I)	-83.94			-117.09					

TABLE 2. Listing of groups and group values — Continued

Group	$\Delta_f H^\circ$ (gas)	C_p° (gas)	S° (gas)	$\Delta_f H^\circ$ (liq)	C_p° (liq)	S° (liq)	$\Delta_f H^\circ$ (solid)	C_p° (solid)	S° (solid)
CHX and CHXO Groups									
C-(H)(C)(CO)(Cl)	-39.88			-35.46	49.45				
C-(C)(CO)(Cl) ₂					74.22				
<i>ortho</i> corr-(I)(I)	7.56	0.00	0.00	6.96	0.00	0.00	5.50	0.00	0.00
<i>ortho</i> corr-(F)(F)	20.90	0.00	0.00	25.00	0.00	0.00	25.50	0.00	0.00
<i>ortho</i> corr-(Cl)(Cl)	9.50	0.00	0.00	14.00	0.00	0.00	8.50	0.00	0.00
<i>ortho</i> corr-(alkyl)(X)	2.51	0.00	0.00	6.30	0.00	0.00	0.00	0.00	0.00
<i>cis</i> corr-(Cl)(Cl)	-4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>cis</i> corr-(CH ₃)(Br)	-4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>ortho</i> corr-(F)(Cl)	13.50	0.00	0.00	18.50	0.00	0.00	19.50	0.00	0.00
<i>ortho</i> corr-(F)(Br)	37.25	0.00	0.00	40.60	0.00	0.00	42.50	0.00	0.00
<i>ortho</i> corr-(F)(I)	85.40	0.00	0.00	83.55	0.00	0.00	85.20	0.00	0.00
<i>meta</i> corr-(I)(I)	0.00	0.00	0.00	0.00	0.00	0.00	20.08	0.00	0.00
<i>meta</i> corr-(COCl)(COCl)	0.00	0.00	0.00	0.00	0.00	0.00	16.06	0.00	0.00
<i>ortho</i> corr-(COCl)(COCl)	0.00	0.00	0.00	0.00	10.58	0.00		0.00	0.00
<i>ortho</i> corr-(F)(CF ₃)	111.00	0.00	0.00	112.00	0.00	0.00	0.00	0.00	0.00
<i>meta</i> corr-(F)(CF ₃)	2.00	0.00	0.00	6.00	0.00	0.00	0.00	0.00	0.00
<i>ortho</i> corr-(F)(CH ₃)	-3.30	0.00	0.00	-6.00	0.00	0.00	0.00	0.00	0.00
<i>ortho</i> corr-(F)(F')	8.00	0.00	0.00	8.00	0.00	0.00	8.00	0.00	0.00
<i>ortho</i> corr-(Cl)(Cl')	8.00	0.00	0.00	8.00	0.00	0.00	8.00	0.00	0.00
<i>meta</i> corr-(F)(F)	0.00	0.00	0.00	6.00	0.00	0.00	8.50	0.00	0.00
<i>meta</i> corr-(Cl)(Cl)	-5.00	0.00	0.00	10.00	0.00	0.00	4.00	0.00	0.00
<i>ortho</i> corr-(Cl)(CHO)	-6.75	0.00	0.00	8.50	0.00	0.00	0.00	0.00	0.00
<i>ortho</i> corr-(F)(COOH)	20.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00	0.00
<i>ortho</i> corr-(Cl)(COCl)	0.00	0.00	0.00	34.43	0.00	0.00	0.00	0.00	0.00
<i>ortho</i> corr-(F)(OH)	25.50	0.00	0.00	23.00	0.00	0.00	20.00	0.00	0.00
<i>ortho</i> corr-(Cl)(COOH)	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00	0.00
<i>ortho</i> corr-(Br)(COOH)	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00	0.00
<i>ortho</i> corr-(I)(COOH)	0.00	0.00	0.00	0.00	0.00	0.00	20.00	0.00	0.00
<i>ortho</i> corr-(NH ₂)(NH ₂)	-10.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
<i>meta</i> corr-(NH ₂)(NH ₂)	0.00	0.00	0.00	0.00	0.00	0.00	14.00	0.00	0.00
<i>ortho</i> corr-(OH)(Cl)	7.50	0.00	0.00	0.00	0.00	0.00	11.00	0.00	0.00
<i>cis</i> corr-(CH ₃)(I)	-4.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Group identities

C-(H) ₃ (C)	=	C-(H) ₃ (C _d) = C-(H) ₃ (C _i) = C-(H) ₃ (C _B) = C-(H) ₃ (O)
	=	C-(H) ₃ (CO) = C-(H) ₃ (N) = C-(H) ₃ (N _A) = C-(H) ₃ (N _I)
	=	C-(H) ₃ (S) = C-(H) ₃ (S) = C-(H) ₃ (SO ₂)
C-(H)(C ₃)	=	C-(H)(C)(C _d) ₂
C _d -(H)	=	C _d -(H)(O) = C _d -(H)(S)
C _d -(H)(C _d)	=	C _d -(H)(C _i) = C _d -(H)(C _B) = C _d -(O)(C _d)
C _B -(H)(C _B) ₂	=	C _B -(H)(O)(C _B) = C _B -(H)(N)(C _B) = C _B -(H)(N _I)(C _B)
	=	C _B -(H)(S)(C _B) = C _B -(C)(O)(C _B) = C _B -(C)(N)(C _B)
	=	C _B -(C)(S)(C _B)
C _B -(C _d)(C _B) ₂	=	C _B -(C _i)(C _B) ₂
C _B -(SO)(C _B) ₂	=	C _B -(SO ₂)(C _B) ₂
S-(C _d) ₂	=	S-(C _B) ₂

TABLE 3. General definitions and examples of notations for organic groups

$C-(H)_3(C)$	A carbon atom with three bonds to hydrogen atoms and the fourth bond to a carbon atom. Example: Ethane.
$C-(H)_2(C)_2$	A carbon atom with two bonds to hydrogen atoms and two bonds to carbon atoms. Example: <i>n</i> -Hexane.
$C-(H)(C)_3$	A carbon atom with one bond to a hydrogen atom and three bonds to carbon atoms. Example: 2-Methylpropane.
$C-(C)_4$	A carbon atom with four bonds to carbon atoms. Example: 2,2-Dimethylpropane.
$C_d-(H)_2$	A doubly-bonded carbon atom attached to two hydrogen atoms. Example: Ethylene.
$C_d-(C)_2$	A doubly-bonded carbon atom attached to two carbon atoms. Example: Propene.
$C_t-(H)$	A triply-bonded carbon atom attached to a hydrogen atom. Example: Ethyne.
$C_t-(C)$	A triply-bonded carbon atom attached to a carbon atom. Example: Propyne.
$C_B-(H)(C_B)_2$	An aromatic ring (benzene) carbon atom bonded to a hydrogen atom and two other aromatic ring carbon atoms. Example: Benzene.
$C_B-(C_B)_3$	An aromatic ring (benzene) carbon atom bonded to three aromatic ring carbon atoms. Example: Biphenyl.
$C_{BF}-(C_{BF})(C_B)_2$	A fused aromatic ring carbon atom (such as the two fused ring carbon atoms in naphthalene) bonded to one other fused aromatic ring carbon atom and aromatic ring carbon atoms. Example: Naphthalene.
$C_{BF}-(C_{BF})_3$	A fused aromatic ring carbon atom bonded to three other fused aromatic ring carbon atoms. Example: Pyrene.
C_a	An allenic carbon atom. When allene is unsubstituted, the group values are equal to allene itself. Example: Allene.
$-CH_3$ corr (tertiary)	A correction for the attachment of each methyl group to a tertiary carbon atom. Example: 2-Methylpropane.
$-CH_3$ corr (quaternary)	A correction for the attachment of each methyl group to a quaternary carbon atom. Example: 2,2-Dimethylpropane.
$-CH_3$ corr (tert/quat)	A correction for the attachment of each methyl group when there is both a tertiary and a quaternary carbon atom present in the longest chain of a hydrocarbon. Example: 2,2,3-Trimethylpentane.
$-CH_3$ corr (quat/quat)	A correction for the attachment of each methyl group when there are two quaternary carbon atoms present in the longest chain of a hydrocarbon. Example: 2,2,4,4-Tetramethylpentane.
<i>ortho</i> corr, hydrocarbons	An aromatic ring correction for <i>ortho</i> substitution in hydrocarbon compounds. Example: <i>o</i> -Xylene.
<i>meta</i> corr, hydrocarbons	An aromatic ring correction for <i>meta</i> substitution in hydrocarbon compounds. Example: <i>m</i> -Xylene.
rsc	Ring strain correction, rsc, for a cyclic non-aromatic compound. Example: Cyclopropane.
rsc (unsub)	Ring strain correction, rsc, for a cyclic non-aromatic unsubstituted compound. Example: Cyclopentane.
rsc (sub)	Ring strain correction, rsc, for a cyclic non-aromatic substituted compound. Example: Methylcyclopentane.

TABLE 3. General definitions and examples of notations for organic groups — Continued

$C-(H)_2(C)(O)$	A carbon atom bonded to two hydrogen atoms, a carbon atom, and an oxygen atom. Example: Methanol.
$O-(C)_2$	An oxygen atom bonded to two carbon atoms. Example: Dimethyl ether.
$C-(H)(O)(C)_2$ (alcohols, peroxides)	Tertiary carbon atom group in alcohols and peroxides. Example: 2-Propanol, <i>n</i> -Heptyl-2-hydroperoxide.
$C-(H)(O)(C)_2$ (ethers, esters)	Tertiary carbon atom group in ethers and esters. Example: Methylisopropyl ether, Isopropyl acetate.
$C-(O)(C)_3$ (alcohols, peroxides)	Quaternary carbon atom group in alcohols and peroxides. Example: <i>tert</i> -Butyl alcohol, <i>Di-tert</i> -butyl peroxide.
$C-(O)(C)_3$ (ethers, esters)	Quaternary carbon atom group in ethers and esters. Example: <i>Di-tert</i> -butyl ether, <i>tert</i> -Butyl acetate.
$C-(H)_2(C)(CN)$	A carbon atom bonded to two hydrogen atoms, a carbon atom, and a nitrile (cyano) group. Example: Propanenitrile.
$CB-(NO_2)(CB)_2$	An aromatic ring carbon atom bonded to a nitro group and two other aromatic ring carbon atoms. Example: Nitrobenzene.
NO_2-NO_2 (<i>ortho</i> corr)	A correction for adjacent (<i>ortho</i>) substitution of NO_2 groups on an aromatic ring. Example: <i>o</i> -Dinitrobenzene.
NO_2-COOH (<i>ortho</i> corr)	A correction for substitution of an NO_2 group adjacent to a $COOH$ group on an aromatic ring. Example: <i>o</i> -Nitrobenzoic acid.
$N-(H)_2(C)$ (first, amino acids)	The first (and only) NH_2 group bonded to a carbon atom in an amino acid. Example: Glycine
$N-(H)_2(C)$ (second, amino acids)	The second NH_2 group bonded to a carbon atom in an amino acid. Example: Lysine
$N-(H)_2(CO)$ (amides, ureas)	A NH_2 group bonded to a carbonyl group, CO , in amides and ureas. Example: Acetamide, Urea.
$N-(H)_2(CO)$ (amino acids)	A NH_2 group bonded to a carbonyl group, CO , in amino acids. Example: Asparagine
$N-(H)(C)(CO)$ (amides, ureas)	A NH group bonded to a hydrogen atom, carbon atom, and a carbonyl group in amides and ureas. Example: <i>N</i> -Methylformamide, Methylurea.
$N-(H)(C)(CO)$ (amino acids)	A NH group bonded to a hydrogen atom, carbon atom, and a carbonyl group in amino acids. Example: Glycylglycine.
Zwitterion energy, aliphatic	A correction for the conversion of an amino acid or to a zwitterion in amino acids and peptides with aliphatic moieties. Example: Glycine, Glycylalanine.
Zwitterion energy, aromatic I	A correction for the conversion of an aromatic amino acid or peptide to a zwitterion containing an aromatic ring attached directly to a conjugation deterring group (such as a $-CH_2-$ group). Example: Phenylalanine, Glycylphenylalanine.
Zwitterion energy, aromatic II	A correction for the conversion of an aromatic amino acid or peptide to a zwitterion containing an aromatic ring attached directly to a conjugation enhancing group (such as a $>C=O$ group). Example: Hippuric acid, Hippurylglycine.

TABLE 3. General definitions and examples of notations for organic groups — Continued

$N_A-(C)$	A doubly-bonded (azo) nitrogen atom bonded to a carbon atom. Example: Azomethane.
$N_A-(C_B)$	A doubly-bonded (azo) nitrogen atom bonded to an aromatic ring carbon atom. Example: <i>trans</i> -Azobenzene.
$N_A-(oxide)(C)$	A doubly-bonded (azoxy) nitrogen atom bonded to a carbon atom. Example: <i>Di-tert</i> -butyldiazene <i>N</i> -oxide
$N_I-(C)$	A doubly-bonded (imino) nitrogen atom bonded to a carbon atom. Example: <i>N</i> -Butylisobutyleneimine.
$N_I-(C_B)$	A doubly-bonded (pyridine-type) nitrogen atom bonded to an aromatic ring carbon atom. Example: Pyridine.
$N_I-(CH_3)$ (<i>ortho</i> corr)	A doubly-bonded (pyridine-type) nitrogen atom in an aromatic ring adjacent to a substituted methyl group. Example: 2-Picoline
N_I-N_I (<i>ortho</i> corr)	A doubly-bonded (pyridine-type) nitrogen atom adjacent to an identical (pyridine-type) nitrogen atom in an aromatic ring. Example: Pyridazine
$C-(H)_2(C)(S)$	A carbon atom bonded to two hydrogen atoms, a carbon atom, and a sulfur atom. Example: Methanethiol.
$S-(C)(S)$	A sulfur atom bonded to a carbon atom and another sulfur atom. Example: Dimethyl disulfide.
$C-(H)_2(C)(F)$	A carbon atom bonded to two hydrogen atoms, a carbon atom, and a fluorine atom. Example: Fluoroethane.
<i>ortho</i> corr, (F)(F)	A correction for the adjacent (<i>ortho</i>) substitution of two fluorine atoms on an aromatic ring. Example: <i>o</i> -Difluorobenzene.
<i>ortho</i> corr, (I)(COOH)	A correction for the substitution of a iodine atom adjacent (<i>ortho</i>) to a COOH group on an aromatic ring. Example: 2-iodobenzoic acid.
<i>ortho</i> corr (Cl)(Cl')	A correction for the substitution of a chlorine atom in an aromatic ring in the near proximity of another chlorine atom in a different aromatic ring which is bonded to the first ring. Example: 2,2'-Dichlorobiphenyl

TABLE 55. Summary of residuals for
C-H-N-O-S-Halogen families

This table provides information on how well agreement was achieved between literature and estimated values for values of $\Delta_f H^\circ$ (in kJ/mol), C_p° , and S° (in J/mol·K). Residuals having $< \pm 4$ indicate good agreement, those between $> \pm 4$ and $< \pm 8$ indicate agreement in the range from fair to just acceptable, and those $> \pm 8$ suggest problems such as poor experimental data, a poor choice of group value, an unaccounted for molecular interaction, or combinations of these problems. The distribution of residuals between gas to condensed phase is about half and half. Compounds which include a ring strain correction, rsc, (such as, "cyclohexane rsc") in their molecular description and compounds which are identified by a single group, such as, methane, formaldehyde, acetonitrile, methyl bromide, etc., are excluded from this tabulation because they have zero residuals. Also excluded are compounds containing a group value in their structural group representation which was derived from a single source of thermodynamic data because such compounds will produce zero residuals. The summary of residuals is divided among the various organic families, and then summed for CH, CHO, CHN, CHNO, CHS, and organic halogen compounds as well as for all families of compounds.

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CH Compounds			
<i>n</i> -Alkanes			
< ± 4	41	35	35
> ± 4 to < ± 8	1	0	0
> ± 8	0	5	5
total	42	40	40
<i>i</i> -Alkanes			
< ± 4	35	46	34
> ± 4 to < ± 8	5	2	9
> ± 8	2	1	5
total	42	49	48
<i>q</i> -Alkanes			
< ± 4	27	22	13
> ± 4 to < ± 8	3	4	9
> ± 8	0	0	2
total	30	26	24
<i>n</i> -Alkenes			
< ± 4	45	34	35
> ± 4 to < ± 8	3	2	1
> ± 8	0	0	0
total	48	36	36
<i>s</i> -Alkenes			
< ± 4	36	17	20
> ± 4 to < ± 8	16	7	5
> ± 8	13	2	1
total	65	26	26
Alkynes			
< ± 4	22	14	13
> ± 4 to	3	0	0
> ± 8	0	0	0
total	25	14	13

TABLE 55. Summary of residuals for
C-H-N-O-S-Halogen families — Continued

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CH Compounds (Continued)			
Alkynes			
< ± 4	22	14	13
> ± 4 to	3	0	0
> ± 8	0	0	0
total	25	14	13
Aromat CH-01			
< ± 4	54	41	30
> ± 4 to < ± 8	6	3	10
> ± 8	5	2	6
total	65	46	46
Aromat CH 02			
< ± 4	56	54	44
> ± 4 to < ± 8	15	7	11
> ± 8	16	10	6
total	87	71	64
Cyclic CH-01			
< ± 4	11	12	10
> ± 4 to < ± 8	6	2	4
> ± 8	4	3	1
total	21	17	15
Cyclic CH-02			
< ± 4	33	32	23
> ± 4 to < ± 8	14	3	4
> ± 8	7	1	2
total	54	36	29
Cyclic CH-03			
< ± 4	15	0	0
> ± 4 to < ± 8	18	0	0
> ± 8	20	0	0
total	53	0	0
Total CH cpds	$\Delta_f H^\circ$	C_p°	S°
< ± 4	375	307	257
> ± 4 to < ± 8	90	30	53
> ± 8	67	24	28
total	532	361	338
CHO Compounds			
Alcohols			
< ± 4	94	56	45
> ± 4 to < ± 8	30	5	6
> ± 8	19	13	5
total	143	74	55
Ethers			
< ± 4	56	25	14
> ± 4 to < ± 8	10	8	11
> ± 8	11	1	0
total	77	34	25

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families - Continued

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CHO Compounds (Continued)			
Aldehydes			
< ± 4	12	10	10
> ± 4 to < ± 8	5	0	2
> ± 8	0	7	5
total	17	17	17
Ketones			
< ± 4	43	14	9
> ± 4 to < ± 8	4	3	3
> ± 8	1	0	2
total	48	17	14
Acids			
< ± 4	68	16	11
> ± 4 to < ± 8	25	9	0
> ± 8	43	3	0
total	136	28	11
Anhydrides			
< ± 4	11	2	1
> ± 4 to < ± 8	3	0	0
> ± 8	4	0	0
total	15	2	1
Esters			
< ± 4	53	21	1
> ± 4 to < ± 8	21	6	0
> ± 8	26	3	3
total	100	30	4
Peroxides			
< ± 4	7	0	0
> ± 4 to < ± 8	0	0	0
> ± 8	3	0	0
total	10	0	0
Hydroperoxides			
< ± 4	4	0	0
> ± 4 to < ± 8	3	0	0
> ± 8	4	0	0
total	11	0	0
Peroxyacids			
< ± 4	2	0	0
> ± 4 to < ± 8	1	0	0
> ± 8	5	0	0
total	8	0	0
Carbonates			
< ± 4	2	1	1
> ± 4 to < ± 8	3	0	0
> ± 8	0	0	0
total	5	1	1
Total CHO cpds	$\Delta_f H^\circ$	C_p°	S°
< ± 4	349	145	92
> ± 4 to < ± 8	105	31	21
> ± 8	116	27	15
total	570	203	128

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families - Continued

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CHN Compounds			
Amines			
< ± 4	67	26	11
> ± 4 to < ± 8	6	3	5
> ± 8	6	3	3
total	79	32	19
Imines			
< ± 4	2	0	0
> ± 4 to < ± 8	1	0	0
> ± 8	0	0	0
total	3	0	0
Nitriles			
< ± 4	31	11	8
< ± 8	4	0	1
> ± 4 to > ± 8	5	1	0
total	40	12	9
Hydrazines			
< ± 4	12	4	4
> ± 4 to < ± 8	0	0	0
> ± 8	0	0	0
total	12	4	4
Diazenes			
< ± 4	14	0	0
> ± 4 to < ± 8	5	0	0
> ± 8	1	0	0
total	20	0	0
Azides			
< ± 4	9	0	0
> ± 4 to < ± 8	0	0	0
> ± 8	0	0	0
total	9	0	0
Cyclic CHN			
< ± 4	32	9	7
> ± 4 to < ± 8	3	1	0
> ± 8	1	1	0
total	36	11	7
Total CHN cpds	$\Delta_f H^\circ$	C_p°	S°
< ± 4	167	50	30
> ± 4 to < ± 8	19	4	6
> ± 8	13	5	3
total	199	59	39
CHNO Compounds			
Amides			
< ± 4	22	12	1
> ± 4 to < ± 8	3	3	0
> ± 8	11	0	1
total	36	15	2

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families - Continued

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CHNO Compounds (Continued)			
Ureas			
< ± 4	23	2	2
> ± 4 to < ± 8	6	0	0
> ± 8	13	0	0
total	42	2	2
Amino Acids			
< ± 4	28	16	5
> ± 4 to < ± 8	13	0	1
> ± 8	5	6	5
total	46	22	11
Nitroso			
< ± 4	8	0	0
> ± 4 to < ± 8	0	0	0
> ± 8	0	0	0
total	8	0	0
Nitro			
< ± 4	65	15	6
< ± 8	5	1	0
> ± 8	23	2	0
total	93	18	6
Nitrites and nitrates			
< ± 4	15	7	6
> ± 4 to < ± 8	2	0	2
> ± 8	0	1	0
total	17	8	8
Nitramines			
< ± 4	13	0	0
> ± 4 to < ± 8	0	0	0
> ± 8	1	0	0
total	14	0	0
Cyclic CHNO (Imides)			
< ± 4	0	0	0
> ± 4 to < ± 8	0	0	0
> ± 8	0	0	0
Total	0	0	0
Total CHNO cpds	$\Delta_f H^\circ$	C_p°	S°
< ± 4	174	52	20
> ± 4 to < ± 8	29	4	3
> ± 8	53	9	6
total	256	65	29
CHS Compounds			
Thiols			
< ± 4	50	29	30
> ± 4 to < ± 8	2	2	0
> ± 8	0	0	2
total	52	31	32

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families - Continued

Family & residual range	Properties		
	$\Delta_f H^\circ$	C_p°	S°
CHS Compounds (Continued)			
Sulfides			
< ± 4	52	31	28
> ± 4 to < ± 8	3	2	1
> ± 8	0	0	3
total	55	33	32
Disulfides			
< ± 4	13	10	8
> ± 4 to < ± 8	1	0	3
> ± 8	0	1	0
total	14	11	11
Sulfoxides			
< ± 4	5	2	2
> ± 4 to < ± 8	2	0	0
> ± 8	1	0	0
total	8	2	2
Sulfones			
< ± 4	27	2	2
> ± 4 to < ± 8	15	0	0
> ± 8	10	0	0
total	52	2	2
Sulfites and sulfates			
< ± 4	5	0	0
> ± 4 to < ± 8	3	0	0
> ± 8	1	0	0
total	9	0	0
Cyclic CHS			
< ± 4	6	3	3
> ± 4 to < ± 8	2	2	1
> ± 8	0	0	1
total	8	5	5
Total CHS	$\Delta_f H^\circ$	C_p°	S°
< ± 4	158	77	73
> ± 4 to < ± 8	28	6	6
> ± 8	12	1	5
total	198	84	84
Halogens			
Fluorides			
< ± 4	30	19	14
> ± 4 to < ± 8	15	5	2
> ± 8	17	0	6
total	62	24	22
Chlorides			
< ± 4	90	49	34
> ± 4 to < ± 8	23	7	4
> ± 8	72	5	3
total	185	61	41

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families — Continued

Halogens (Continued)			
Bromides			
< ± 4	44	22	9
> ± 4 to < ± 8	9	2	6
> ± 8	6	9	1
total	59	33	16
Iodides			
< ± 4	26	10	7
> ± 4 to < ± 8	13	1	1
> ± 8	8	2	1
total	47	13	9
Mixed halogens			
< ± 4	10	7	4
> ± 4 to < ± 8	4	1	3
> ± 8	2	2	2
total	16	10	9
Total halogens	$\Delta_f H^\circ$	C_p°	S°
< ± 4	200	107	68
> ± 4 to < ± 8	64	16	16
> ± 8	105	18	13
total	369	141	97

TABLE 55. Summary of Residuals for
C-H-N-O-S-Halogen Families — Continued

All compounds	$\Delta_f H^\circ$ (%)	C_p° (%)	S° (%)
< ± 4	1423 (67)	738 (80)	540 (76)
> ± 4 to < ± 8	335 (16)	91 (10)	105 (14)
> ± 8	366 (17)	84 (10)	70 (10)
Grand total	2124 (100)	913 (100)	715 (100)